A Daily Return-Based Multi-Asset Factor Model

Construction, Estimation and Out-of-Sample Validation

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Abstract. This paper documents a factor model that estimates exposures from observed returns alone, without holdings data or look-through. Forty daily factors are constructed across nine blocks — equity market, equity style, rates, credit, foreign exchange, commodities, volatility, liquidity and stress, and alternative risk premia — from 254 instruments and 51 level series, giving 200,958 factor observations from December 2003 to October 2026. Every model input is a return, and stationarity is tested rather than assumed: a six-test battery gates the panel, with the Dickey-Fuller and KPSS tests read jointly because neither alone identifies the failure that matters. The factor panel is maintained in two forms, raw and block-orthogonalised, and the model can be estimated end to end on either, which turns the case for orthogonalisation from an in-sample argument into an out-of-sample measurement: on a representative security the raw panel fits better in sample (adjusted R^2 of 0.481 against 0.441) and forecasts distinctly worse out of it (Mincer-Zarnowitz slope 0.33 against 0.93). Residualisation is fitted causally on a trailing window, at a measured cost of -0.05 residual correlation rather than the exact zero a full-sample fit would give. Risk forecasts are scored, not asserted: bias statistics, Mincer-Zarnowitz regressions and both the Kupiec and Christoffersen coverage tests are run on every specification, every forecast is compared with the security's own EWMA volatility, and forecasts built on ill-conditioned designs are flagged and excluded rather than silently used. Since October 2026 the specification is chosen by a model agent that searches it on a validation span and judges the winner on a held-out one. Eleven sector factors, a lead-lag adjustment of the multi-day factor covariance and a portfolio-level validation over 54 large caps test the part of the model a single security cannot: whether its risk aggregates. An appendix states every statistical procedure in the form in which it is implemented.

Implementation: `bastofferm/factor-terminal`. Python 3.12, PostgreSQL 16, FastAPI, Next.js. 517 tests.

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1 Introduction

1.1 The question the model answers

A holdings-based factor model asks what a portfolio owns and maps those holdings onto factor exposures. It is the better method when holdings are available, current and complete. For a great deal of what an analyst actually has to price, they are none of these: a mutual fund discloses quarterly and with a lag, a hedge fund may not disclose at all, and a systematic strategy can turn over its book faster than any disclosure cycle. Look-through into a fund of funds compounds the problem at every level.

The model documented here asks a different question — not *what does this security own*, but *what does its return actually move with* — and answers it from the return series alone. This is the return-based tradition of Sharpe (1992), applied across asset classes at daily frequency and with the statistical machinery that daily data requires.

The advantage is availability: any security the data warehouse can price can be estimated, which in the present installation means 9,434 of them. The cost is that the model sees only what has appeared in returns. A risk taken but not yet realised — a concentrated position that has not moved, an option position whose gamma has not been struck, a credit exposure that has not defaulted — is invisible. Section 9 states this and the other limits plainly, and the specific-risk floor of Section 6.3 is the model's own structural admission of it.

1.2 What a return-based model must be defended against

Estimating exposures from returns means estimating realised sensitivities, and a realised sensitivity is a sample statistic with all of a sample statistic's ways of going wrong. Four failure modes recur, and most of the engineering in this project exists to close them:

Spurious correlation.
Two series can be strongly related over a sample for no economic reason, particularly when both are non-stationary. The defence is to admit only stationary return series (Appendix A.2) and to reduce the number of ways a factor can stand in for another (Section 4).

Stale pricing.
An instrument that does not trade every day, or whose price is smoothed by an administrator, produces autocorrelated returns and understated betas. Two tests in the battery exist specifically to detect it: the Lo-MacKinlay variance ratio and the Ljung-Box statistic. The Dimson correction of Section 5.1 is the remedy where the diagnosis is confirmed.

Non-stationarity.
A level series used as a regressor — a yield, a spread, an index — makes every standard error in the regression meaningless. No level enters this model; every input is transformed to a return first, and the transformation is part of the factor's definition.

Short histories.
Style factor proxies begin in 2011–2013 and the funding liquidity factor in 2018. A 756-day window over a factor with 400 days of history silently drops it. The model records the usable factor count on every window rather than reporting a loading vector whose basis has quietly changed.

1.3 What this document contains

Sections 2 through 6 describe the model as it is implemented: the data and its audit, the factor universe, orthogonalisation, exposure estimation, and covariance and risk forecasting. Section 7 is the operating manual — what runs, in what order, and how the data is kept current — and Section 8 describes the software that carries it. The latter is not incidental: the decision to keep every statistical routine as a pure function over arrays is what makes the accuracy claims in this paper checkable, and it is the single most consequential design choice in the project. Section 9 states the limits.

Appendix A gives each statistical procedure in the form in which it is implemented, including the conventions — bandwidth rules, degrees of freedom, boundary handling — that decide whether two nominally identical implementations agree. Appendix B lists the factor registry.

Every figure quoted in the text was measured on the panel described in Table 1 rather than taken from the literature. Where a figure is unflattering it is reported anyway; a validation section that reports only successes is not a validation section.

Table 1. The panel, as of 7 October 2026. Sector factors excluded.

Factors	40	Factor observations	200,958
Blocks	9	First observation	2003-12-02
Priced instruments	254	Last observation	2026-10-07
Level series	51	Securities estimable	9,434

1.4 Notation

Throughout, $r_{i,t}$ is the log return of security i on day t, $r_{f,t}$ the cash rate for that day, and $F_t \in \mathbb{R}^K$ the vector of factor returns. The model is

$$r_{i,t} - r_{f,t} \;=\; \alpha_i \;+\; \beta_i' F_t \;+\; \varepsilon_{i,t}, \qquad \mathbb{E}[\varepsilon_{i,t} \mid F_t] = 0, \tag{1}$$

and the risk decomposition that follows from it is

$$\operatorname{Var}(r_p) \;=\; \beta_p' \, \Sigma_F \, \beta_p \;+\; w' \Sigma_\varepsilon \, w, \tag{2}$$

with Σ_F the factor covariance matrix, Σ_ε the (diagonal) specific covariance, and w portfolio weights. All factors are daily log excess returns over the effective federal funds rate (`FRED:DFF`), denominated in US dollars. The annualisation constant is 252 trading days throughout.

Equation (2) carries a constraint that is easy to state and easy to violate in code: Σ_F must be the covariance of exactly the series the βs were estimated on. Section 4.4 returns to it, because the model maintains two factor panels and pairing one panel's betas with the other's covariance produces a silently wrong number.

2 Data

2.1 Sources

Prices and levels come from three places. A PostgreSQL warehouse, reached over `postgres_fdw`, supplies equity prices, fund prices, foreign exchange, and the euro-area and Japanese government curves sourced from the ECB, the Bank of Japan and the Japanese Ministry of Finance. The Federal Reserve Economic Database supplies US Treasury constant-maturity yields, the ICE BofA option-adjusted spread ladder, policy and money-market rates, and the financial-conditions indices. Yahoo Finance supplies exchange-traded fund and index prices where the warehouse lacks them.

Each of the 203 priced instruments carries the provenance of its own series — source system, the original ticker or series identifier under which it is published, and the full history available — and the application displays it next to the chart rather than in a footnote. A user looking at a line should be able to see, without leaving the page, which vendor produced it and under what name.

2.2 The block audit

Nine blocks of daily proxies are required. Before anything was built, each was audited against what was actually obtainable, because a factor built on the wrong series is worse than a missing factor: it looks entirely plausible and it is wrong. Table 2 is the result.

Two gaps could not be closed, and are carried as documented approximations rather than quietly patched:

- **FX carry.** The warehouse holds no forward points, so the carry factor is built from policy-rate differentials across the three non-USD policy rates available. It is narrower and noisier than a G10 carry basket, and Section 4.7 shows it retains a measurable dependence on the yen even after residualisation.
- **Euro-area and Japanese rates.** These arrive through the warehouse, which this project does not itself refresh. They lag by design, and the Data Health page reports the lag rather than hiding it.

2.3 From level to return

Every input is transformed into a return before it can become a factor. Four transformations cover the panel, and each is a modelling decision rather than a formatting step.

Prices. Log differences of a total-return price series. Price-return index levels are excluded: a `^`-prefixed Yahoo index carries no dividends, and using one as an equity factor biases every equity beta down by the dividend yield. Such series are flagged `is_total_return = false` in the registry and are not eligible for construction.

Yields. A yield is $I(1)$ and is never a regressor. A constant-maturity yield becomes the synthetic total return of a par bond at that tenor:

$$r_t \;=\; -D_{\mathrm{mod}}(y_{t-1})\;\Delta y_t \;+\; \tfrac{1}{2}\,C(y_{t-1})\,(\Delta y_t)^2 \;+\; \frac{y_{t-1}}{252}, \tag{3}$$

Table 2. Block audit: what was required, what the data supported, what was done.

Block	Verdict	Action
Rates	Complete US, EA and JP curves, daily	Used in full
Credit	Full ICE BofA OAS ladder (AAA–CCC), plus IG, HY, loan and EM funds	Used in full
Equity style	Style funds, with Fama-French and AQR for validation	Used, with caveat
Volatility	VIX futures funds, not index levels	Used in full
Commodities	Total-return funds	Used; *not* the futures series
Alt. risk premia	Constructible in-house from the return panel	Built rather than sourced
Equity market	**Insufficient** — only price-return index levels	12 total-return funds added
Rates (gilt)	**Missing** — no daily UK curve in the warehouse	`IGLT.L` added as proxy
Liquidity	**Thin** — TED and LIBOR-OIS both discontinued	SOFR, EFFR, OBFR, TGCR, NFCI, ANFCI, STLFSI4 added

with duration and convexity evaluated at *yesterday's* yield. Evaluating them at y_t would price the move using information contained in the move itself. When a cash rate is supplied the carry leg becomes excess carry and the result is an excess return directly.

Spreads. An option-adjusted spread becomes a duration-matched credit excess return,

$$r_t = -D_{\text{spr}}\,\Delta s_t + \frac{s_{t-1}}{252}, \tag{4}$$

which by construction contains no government-rate component. This is what keeps the credit block from double-counting the duration already carried by the rates block.

Low-frequency series. A weekly or quarterly series is never forward-filled onto the daily grid. A filled series carries no news between releases, yet the regression counts every filled day as an independent observation: standard errors are understated and autocorrelation is manufactured. Such series become *release-event* factors instead. The change is standardised by the trailing standard deviation of its own past releases and recorded on the publication day; the factor is exactly zero on every other day, and undefined before the first usable release.

Sparse rather than filled

A release-event factor is zero most of the time, which looks strange on a chart and is correct. The alternative — a step function that changes once a week — is a series whose autocorrelation is an artefact of the fill, and every t-statistic computed against it is overstated.

2.4 The trading calendar

The model runs on a trading calendar derived from the union of observation dates across the instrument panel. Cryptocurrency instruments are excluded from that union, and the reason is worth recording because the failure it caused was silent.

Crypto prices seven days a week. Included in the calendar, it added 1,229 weekend rows on which every other factor is missing. Two consequences followed. A 252-row estimation window then spanned fewer than 252 trading days, so windows were shorter than they claimed to be. And equity factors, missing on every weekend row, fell below the coverage threshold used to admit a factor into the covariance matrix. The defect surfaced only because the matrix endpoint began returning 8 factors where 39 were expected; nothing raised an error, and no individual number looked implausible.

2.5 Liveness, staleness and currency

Three of the instruments in the panel have stopped publishing: `^TYVIX` (discontinued 2020), `^EVZ` (ends March 2025) and `JJC` (ends June 2025). They are flagged dead rather than deleted, because deleting them would silently shorten the history of any factor that used them. A liveness check re-runs nightly and can revive a series that resumes.

Securities pulled on demand are converted to the base currency at ingestion. A Japanese equity arrives in yen; storing its local return and regressing it on dollar-denominated factors makes the yen leg appear as a spurious exposure. The conversion is $r^{\text{USD}} = r^{\text{local}} + r^{\text{fx}}$ in logs, and it is applied before the return is written. The correctness of this

step was established by regression rather than by inspection: regressing the difference between the stored series and the dollar series on the yen return gave a slope of -0.997, which is the signature of exactly one missing currency leg.

3 The factor universe

3.1 Blocks and hierarchy

Forty factors span the nine blocks. Each carries a hierarchy level: level-0 factors are used raw, and factors at higher levels are residualised against the levels above them in the block ordering. Table 3 lists the panel with the annualised volatility range of the *raw* series, before any residualisation.

Table 3. The factor panel. Volatilities are of the raw series.

Block	Factors	Ann. vol
Equity market	`eq_global`, `eq_us`, `eq_europe`, `eq_japan`, `eq_em`, `eq_size`, `eq_cyclical`	10.6–27.3%
Equity style	`sty_value`, `sty_momentum`, `sty_quality`, `sty_lowvol`	13.4–20.2%
Rates	`rt_us_level`, `rt_us_slope`, `rt_us_curve`, `rt_ea_level`, `rt_jp_level`, `rt_uk`, `rt_breakeven`	1.4–12.6%
Credit	`cr_ig`, `cr_hy`, `cr_loans`, `cr_em_hard`, `cr_em_local`	5.8–10.8%
FX	`fx_usd`, `fx_jpy`, `fx_em`, `fx_carry`	7.5–11.5%
Commodities	`cm_broad`, `cm_energy`, `cm_gold`, `cm_agri`, `cm_industrial`	16.7–37.6%
Volatility	`vol_equity`, `vol_rates`, `vol_variance_premium`	10.0–67.7%
Liquidity & stress	`liq_conditions`, `liq_funding`, `liq_risk_off`	10.0–20.8%
Alt. risk premia	`arp_trend`, `arp_xs_momentum`	5.5–19.5%
Equity sector*	`sec_tech`, `sec_financials`, `sec_health`, `sec_industrials`, `sec_discretionary`, `sec_staples`, `sec_energy`, `sec_materials`, `sec_utilities`, `sec_real_estate`, `sec_communication`	14.0–29.4%

*Role *sector*: not among the forty and in no default factor set. A single stock may add the factor of its own GICS sector.

The eleven sector factors are SPDR sector ETFs residualised against `eq_global`, `eq_us` and `eq_cyclical`. They are kept out of the forty because eleven more regressors on every window would cost more in beta noise than they explain; a stock adds only its own sector's, and the model agent decides whether it earns its place. What they buy is not visible in one stock's total risk, which they mostly move from specific to systematic, but in the risk of portfolios (Section 6.8).

3.2 Construction, written out

A factor identifier is not a definition. `cm_broad` could be a futures index, a fund, or a basket; `rt_us_slope` could be a yield difference, a return difference, or a duration-neutral steepener. The companion document *factor-formulas* therefore gives all forty factors as explicit formulas over named instruments, together with the exact equation that orthogonalises each one. It is generated from the registry rather than written by hand, and the transcription is verified rather than trusted: a test evaluates each rendered formula independently and asserts that it reproduces the builder's own output, so a formula in the document cannot drift away from the code that produced the data.

The 10s–2s steepener illustrates why the appendix is necessary. Its registry entry reads `curve`; its definition is

$$f_t = 5 \cdot \left(u_t^{(10y)} - u_t^{(2y)}\right), \qquad u_t^{(T)} = \frac{b_t^{(T)}}{D_{t-1}^{(T)}}, \tag{5}$$

$$b_t^{(T)} = -D_{t-1}^{(T)}\,\Delta y_t^{(T)} + \tfrac{1}{2} C_{t-1}^{(T)} \left(\Delta y_t^{(T)}\right)^2 + \frac{y_{t-1}^{(T)} - c_{t-1}}{252}. \tag{6}$$

Yield changes at the two tenors become bond returns by equation (3); each leg is divided by its own duration so that the two are comparable in units; and the difference is scaled back to a five-year equivalent. None of that is recoverable from the word `curve`, and an analyst who assumed the obvious reading — a difference of yields — would misinterpret every loading on it.

3.3 Validation against published factors

Published factor series — the Fama-French set (Fama and French, 1993) and the AQR library — lag by one to two months and can never drive a daily model. They are kept for one purpose: to establish that a construction measures what its name claims. Table 4 is the strongest single piece of evidence in the project, and it is simultaneously the clearest demonstration of what orthogonalisation is for.

Table 4. Correlation with published reference factors, raw and orthogonalised.

Factor	Reference	Raw	Orthogonalised	Overlap
`eq_global`	Mkt-RF	**0.963**	0.963	4,551
`eq_size`	SMB	0.914	**0.874**	4,287
`sty_momentum`	Mom	0.207	**0.638**	3,006
`sty_value`	HML	0.194	**0.499**	3,006
`sty_lowvol`	BAB (AQR)	−0.002	**0.199**	3,363
`sty_quality`	RMW	−0.145	**0.143**	2,943

The style rows repay careful reading. As raw long-only fund returns, the style proxies correlate with their academic counterparts at 0.19 or below — low volatility at zero, quality negatively — because market beta swamps the style. Removing market and sector raises momentum to 0.64 and value to 0.50. That is orthogonalisation doing precisely the job it exists for, and it is measured rather than asserted.

It also shows the honest limit. Quality reaches 0.14 and low volatility 0.20. These are weak proxies, kept because the block would otherwise be incomplete and not because they are good; Section 9 repeats the warning.

Value against momentum comes out at −0.11 after orthogonalisation, reproducing the well-known negative relationship (Asness et al., 2013) in sign though not in the −0.3 to −0.5 magnitude usually reported for true long-short portfolios. That gap is what a long-only proxy costs.

4 Orthogonalisation

4.1 Why

Regressing a security on `eq_global` and raw `eq_japan` simultaneously is a poor idea. The two are strongly correlated; the design matrix is ill-conditioned; and the estimated betas emerge as large offsetting numbers that are individually meaningless even when their sum is stable. This is not a subtlety — it is the standard consequence of near-collinear regressors — but it is easy to overlook in a forty-factor model where no single pair looks alarming.

The block hierarchy removes the overlap. A factor at level ℓ is replaced by its residual after projection onto the factors at levels below it:

$$\tilde{f}_t^{(\ell)} = f_t^{(\ell)} - \hat{a} - \hat{b}' f_t^{(<\ell)}. \tag{7}$$

The loading on $\tilde{f}^{(\ell)}$ then answers a specific question — what this block adds beyond the blocks above it — rather than an ambiguous one.

4.2 Causality: when the residualisation is fitted

The implementation choice that matters most is not *whether* to residualise but *when* the coefficients $\hat{a}, \hat{b}$ in (7) are estimated.

Full-sample residualisation is exactly orthogonal and wrong. It puts information from the end of the sample into factor values at the beginning, which flatters the model in precisely the historical periods where it is being judged. This implementation fits on a trailing 504-day window, refitted every 21 days, requiring at least 252 observations. The cost of that choice is measurable, and was measured (Table 5).

Table 5. Residual correlation with the regressor block, by fitting scheme.

Scheme	Residual correlation	Lookahead
Rolling, 504 days (used)	−0.05	none
Expanding	−0.34	none
Full sample	0.00	**yes**

A residual correlation of −0.05 instead of a clean zero is the price of not cheating. The expanding window is worse than either, and instructively so: an expanding fit never forgets, so a factor whose loading was high in 2008–09 keeps a high fitted beta for a decade afterwards and the residual acquires a systematic *negative* loading on the regressor. A trailing window tracks a time-varying beta using only past data.

Two years, refitted monthly

Long enough to estimate a handful of loadings with tolerable precision; short enough to follow a regime change. Refitting every 21 days rather than every day is a deliberate smoothing choice — a daily refit makes the residual jump on days when nothing happened to either series except the arrival of one new observation in the fitting window.

4.3 What the orthogonalised series is, and is not

The orthogonalised series is what the regression, the covariance matrix and the risk forecast consume by default. It is emphatically not what the factor earned. `eq_us` returns 12.9% a year at 17.3% volatility; its residual after removing `eq_global` returns −0.2% at 4.2%. Both numbers are correct and they describe different objects.

The application keeps them on separate pages for exactly this reason: the Factor Explorer describes the model, and the Raw Explorer describes the data. A page that mixed them would invite the reading that the model's equity factor lost money over two decades, which is false.

4.4 Both panels, end to end

Orthogonalisation is a modelling choice, not a fact about the world. The model therefore stores every factor twice — `ret_excess` for the raw series, `ret_orth` for the residual — and estimates end to end on either, with the choice travelling as part of the specification:

```
python -m backend.pipeline.run_estimation --instrument US:AAPL --window 252 --step 1m
python -m backend.pipeline.run_estimation --instrument US:AAPL --window 252 --step 1m
    --raw-factors
```

These hash to different specification identifiers, so the two coexist in the cache rather than overwriting one another.

The constraint that makes this more than a display option is equation (2). Portfolio risk is $\beta'\Sigma\beta$, so Σ must be the covariance of the same series the βs refer to. The risk job therefore reads the specification's own panel flag and builds Σ from the matching panel. It did not always: until this was corrected, a specification estimated on raw factors was scored against the orthogonalised covariance, and every predicted volatility it produced was wrong — silently, with no error raised and no implausible number to give it away. It is the exemplary case of the failure mode this project is built to make visible.

Table 6 summarises what each panel is good for; the choice is a matter of the question being asked rather than of one panel being correct.

Table 6. What each panel is good for.

	Raw	**Orthogonalised**
Reading a beta	Total exposure: 1.02 to global equity means what it says	Incremental: what the factor adds beyond the blocks above it
Design matrix	Blocks overlap by construction; high VIFs by definition	Well conditioned — which is what the hierarchy is for
Attribution	Double-counts: high-yield credit carries duration *and* equity	Each block's contribution is its own
Best used for	Explaining an exposure to someone	Decomposing risk, and forecasting it

4.5 What the raw panel costs, measured out of sample

The argument for the hierarchy is usually made from in-sample statistics: correlations, condition numbers, residual correlations. Estimating both panels end to end turns it into an out-of-sample measurement.

Table 7 holds every setting fixed — 252-day window, monthly step, ordinary least squares, blended covariance, 21-day horizon — on a single representative security. Each

panel yields 249 monthly windows, and 5,208 daily observations stand behind the coverage tests.

Table 7. Orthogonalised against raw, identical settings, US:AAPL.

	Orthogonalised	**Raw**
Windows excluded as ill-conditioned	**0** of 248	2 of 248
Mean max VIF (worst)	25 (90)	4,990 (230,726)
Mean condition number, standardised (worst)	16 (41)	117 (1,451)
Mean adjusted R^2 in sample	0.441	**0.481**
Bias ratio (target 1)	**0.901**	0.868
Mincer-Zarnowitz slope (target 1)	**0.93** ($p = 0.69$)	0.33 ($p = 0.0002$)
Mincer-Zarnowitz R^2	**0.156**	0.060
QLIKE against own EWMA	0.322 vs 0.290 ($p = 0.16$)	0.372 vs 0.292 ($p = 0.015$)
VaR 95% breaches	208 of 260 ($p = 0.0006$)	199 of 258 ($p = 0.0001$)

The raw panel fits *better* in sample — adjusted R^2 of 0.481 against 0.441 — and forecasts distinctly worse out of it. That is the entire case for orthogonalisation in two lines, and it is the opposite of what an in-sample fit statistic would tell you.

A Mincer-Zarnowitz slope of 0.33 says that the raw-panel forecast barely tracks the variation in realised risk, which is exactly what unstable betas on a collinear design do; its loss is significantly worse than the security's own EWMA volatility, which the orthogonalised panel's is not.

An earlier version of this table reported a bias ratio of 0.964 and 248 breaches for the orthogonalised panel, with Kupiec passing. Those depended on shrinking specific risk toward a median taken over future windows and over unrelated securities that shared the specification, removed in October 2026. Without it neither panel's VaR is calibrated as built: both over-forecast single-stock risk by about 10%, a third of it estimation noise in the betas.

Both panels fail the Christoffersen test ($p < 0.0001$): breaches cluster. The cause is a forecast held for a month while volatility moves within it — refreshing the covariance daily removes most of it — and it is identical on both panels, so it is not evidence about the panel choice. Reporting it anyway is the point — a validation section that suppressed it would be misrepresenting the model.

The covariance matrix tells the same story from the other end (Table 8). Same 39 factors, same sample — January 2015 to May 2026, 2,794 days — and the same blended estimator:

Table 8. Covariance structure of the two panels.

	Orthogonalised	**Raw**
Condition number	1,068	**9,254**
Smallest eigenvalue	2.19×10^{-4}	7.50×10^{-5}
PC1 share of variance	34.5%	**50.3%**

Half the variance of the raw factor set lies in a single direction, which is another way of saying that a forty-factor model built on it is not really a forty-factor model. The application draws both, because the raw heatmap is the clearest single picture of what

the hierarchy is for.

4.6 A gate that had to be made panel-aware

Forecasts built on ill-conditioned windows are excluded from scoring (Section 6.9). That gate originally tested two conditions — a condition number above 200, or a maximum variance inflation factor above 100 — and the second does not survive contact with the raw panel. On raw factors, `eq_us` regressed on the other thirty-nine has an R^2 near 0.9998, because `eq_global` is one of them. Applied unchanged, the rule excluded 236 of 249 raw-panel windows and left thirteen forecasts to score, which is not an estimable model.

The question is whether those 236 windows deserved exclusion, and the evidence says they did not. Their forecasts averaged 0.34 predicted against 0.28 realised, with a maximum of 0.68 — nowhere near the 3.6 that motivated the gate. Their mean condition number was 147, comfortably inside the limit. They were being rejected for a property of the factor set rather than a defect in the window.

The condition number therefore gates both panels and the VIF limit applies only to the orthogonalised panel, where a VIF above 100 is genuinely anomalous. Since October 2026 the condition number is computed on standardised regressors: on the raw columns it mostly measured the spread of factor volatilities, which run from 1.4% to 68%, and correlated 0.86 with it. VIF is still recorded on every forecast either way, keeping the role it was introduced for: naming *which* factor is responsible.

Recorded, not gated

This is the same treatment the ARCH-LM test receives in the stationarity battery. Where a statistic detects a property of the design rather than a defect in it, the right response is to record it and say so, not to reject on it. Gating on ARCH-LM would exclude essentially every financial series; gating on VIF would exclude the raw panel by construction.

The raw panel still loses 16 windows to the condition-number check, all of them in 2002–03 where few factors yet existed. Relaxing the VIF rule does not make the panel ungated: a window with a condition number of 610 is excluded on either panel.

4.7 Per factor, side by side

The Factor Explorer carries a raw-against-orthogonalised comparison for whichever factor is selected: both cumulative paths on one axis, the difference between them, and what the residualisation achieved — the correlation with each target before and after, and the average loading that produced it. Table 9 reports the measurement on the common sample.

Two rows deserve comment. `fx_carry` still correlates +0.16 with `fx_jpy` after residualisation — the sign has flipped but the magnitude has not gone away, which is what a three-legged basket dominated by one funding currency looks like even after that currency is regressed out. The factor's registry entry already advises treating its loading sceptically, and this is the measurement behind the advice.

And `arp_trend` has *negative* variance removed: the residual is fractionally noisier than the factor it came from. That is not an arithmetic defect but a property of a causal fit. A loading estimated on the trailing two years and applied for the following month adds variance rather than removing it when the true loading is near zero and moving. Both

Table 9. Effect of residualisation, selected factors.

Factor	Variance removed	vs. largest target, before → after
`eq_us`	94.1%	`eq_global` 0.967 → −0.020
`liq_risk_off`	80.9%	`eq_global` −0.847 → −0.003
`cr_em_local`	38.2%	`fx_usd` −0.536 → −0.015
`fx_carry`	38.7%	`fx_jpy` −0.659 → **+0.164**
`arp_trend`	−1.9%	`eq_global` 0.125 → 0.036
`cm_broad`	0.0%	nothing above it; the panels are identical

rows are displayed rather than smoothed over.

5 Exposure estimation

5.1 The regression

Exposures come from rolling regression of an instrument's excess return on the factor panel, in the form of equation (1). Every control that changes the answer is a user choice rather than a hard-coded constant, because each one is a modelling judgement and none has a default that is right for every instrument:

Estimation window	63–756 days
Roll-forward step	1d / 1w / 1m / 3m / 6m
Estimator	OLS, Huber (robust), or ridge
Weighting	Equal or exponentially decayed (half-life in days)
HAC lags	Automatic (Newey-West rule) or explicit
Dimson lead-lag	0–2
Winsorisation	Optional, at the 1st and 99th percentiles
Factor panel	Orthogonalised or raw

Appendix A states each estimator precisely. Three of them deserve a word here about *when* to reach for them.

Huber. A single day — an earnings surprise, an index reconstitution, a halt — can move a 252-day beta materially under least squares. Huber weighting caps the influence of such a day without discarding it, which is usually preferable to winsorisation because it adapts to the scale of the residuals rather than to a fixed quantile.

Ridge. Where the design is genuinely collinear — which the raw panel is by construction — a small ridge penalty trades a little bias for a large reduction in variance. The penalty is applied to standardised regressors and scaled by σ^2; Appendix A.3.3 notes why the opposite scaling, which was the original implementation, silently shrinks every loading to zero.

Dimson. An instrument that prices before the factor close — a Japanese equity against US-close factors, a fund with stale marks — reacts to a factor with a delay. Its contemporaneous beta is then biased toward zero. The Dimson correction (Dimson, 1979), in the

tradition of Scholes and Williams (1977), adds lagged factor returns to the design and sums the coefficients:

$$r_{i,t} = \alpha + \sum_{\ell=0}^{L} \beta^{(\ell)\prime} F_{t-\ell} + \varepsilon_t, \qquad \hat{\beta}^{\text{Dimson}} = \sum_{\ell=0}^{L} \hat{\beta}^{(\ell)}. \tag{8}$$

Only lags are included, not leads: a lead would use tomorrow's factor return to explain today's security return, which is unavailable at forecast time and would make the model non-causal. The standard error of the sum accounts for the covariance between the lag coefficients, which are typically negatively correlated; adding variances alone — the conservative fallback when no covariance is supplied — overstates the uncertainty.

5.2 Per-window diagnostics

Every window stores its own diagnostics rather than reporting only the loading vector: R^2 and adjusted R^2, the F statistic and its p-value, root mean squared error, annualised residual volatility, Durbin-Watson, the design condition number, the maximum variance inflation factor, Ljung-Box and ARCH-LM on the residuals, and the change in the loading vector since the previous window.

These are not decoration. The condition number and the maximum VIF are what the reliability gate of Section 6.9 reads; Ljung-Box and ARCH-LM on the residuals are what justify HAC standard errors; and the residual volatility is the input to specific risk.

5.3 Stability

A loading vector that jumps between windows is a drift alert: the exposure the model reports is no longer the exposure it reported last month, and any risk number built on it inherits that instability. Two measures are kept — the L_1 change in the whole vector, and its correlation with the previous window.

The comparison is made on the factors *common to both windows*, with the size of that common set recorded alongside. An earlier version refused to compare whenever the usable factor set changed at all, which put gaps in the chart at exactly the moments the exposure set moved — 169 of 178 points missing — which is when drift is most worth seeing. Since the L_1 sum falls mechanically as the basis narrows, the overlap count must be read with it.

5.4 Specification hashing

Every parameter combination — window, step, estimator, weighting, half-life, ridge penalty, Dimson lags, winsorisation, HAC lags, factor panel — hashes to a `spec_id`. Loadings, diagnostics, covariance matrices and risk forecasts are all keyed by it.

Two things follow. Re-requesting a view an analyst has already seen is a table read rather than several hundred regressions. And a result is reproducible by construction: the specification that produced any stored number can be recovered from its identifier and re-run. The application exposes this directly — the run inspector unpacks a stored `spec_id` into the parameters that produced it, so a number on screen can always be traced back to the settings behind it.

6 Covariance and risk forecasting

6.1 Factor covariance

The factor covariance is estimated by one of four methods — sample, exponentially weighted, Ledoit-Wolf shrinkage to a constant-correlation target, or a blend of the last two:

$$\Sigma_F \;=\; a\,\Sigma_{\text{EWMA}} \;+\; (1-a)\,\Sigma_{\text{shrunk}}, \qquad \Sigma_r \;=\; B\,\Sigma_F\,B' \;+\; \Sigma_\varepsilon. \tag{9}$$

Shrinkage is not optional at this dimension. With 39 factors on a rolling window, the sample covariance is badly conditioned, and an optimiser handed an indefinite matrix will happily construct a portfolio out of the negative eigenvalue. The constant-correlation target of Ledoit and Wolf (2003) keeps each factor's own variance and replaces every correlation by the average correlation, which is a considerably better description of a factor panel than the identity target used in the simpler version of the method.

6.2 Positive semi-definiteness

Positive semi-definiteness is checked on every estimate and, where it fails, repaired by eigenvalue clipping followed by a rescale that restores the original factor volatilities. The rescale matters: clipping alone inflates the diagonal, which shifts every risk number computed from the matrix. The repair is recorded on the result rather than applied silently.

6.3 Specific risk

Idiosyncratic volatility is

$$\sigma_\varepsilon^2 \;=\; \max\!\left\{\,\sigma_{\text{floor}}^2,\; \hat{\sigma}_\varepsilon^2\,\right\}, \qquad \sigma_{\text{floor}} = 2\%\ \text{p.a.} \tag{10}$$

where $\hat{\sigma}_\varepsilon^2$ is the loading window's residual variance or, as an option the model agent searches, an EWMA of the out-of-sample residuals of the betas in force, each corrected by $f'Vf$ for the beta noise it contains. The floor is the model's structural admission that it cannot see every risk a security carries: without it, an optimiser would load up on an instrument whose risk merely *looks* tiny. An earlier version shrank the estimate toward a peer median that was taken over future windows and unrelated securities; it was removed.

The residual is not noise to be discarded. For an active fund it holds manager decisions, fees and unmodelled derivatives, and it is frequently the most interesting part of the decomposition.

6.4 The forecast and its lag

For a forecast dated t, the loadings come from a window ending at or before t and the covariance from returns up to t; the realised volatility it is compared against covers $t+1,\ldots,t+h$. The lag is enforced structurally in the query that assembles the comparison, not by convention or by a comment, because a validation framework whose lookahead protection depends on a developer remembering it will eventually fail.

6.5 Scoring

Four tests, each answering a different question:

Bias statistic.
The standard deviation of returns divided by the forecast in force at each date. Target 1; above 1 means risk was underestimated. It is the sharper statistic than the mean ratio of realised to predicted volatility, which is right-skewed and noisy.

Mincer-Zarnowitz.
Realised variance regressed on predicted variance, with a joint Wald test of $(\alpha, \beta) = (0, 1)$ (Mincer and Zarnowitz, 1969). Run on variances rather than volatilities because variance is what the model forecasts and what aggregates linearly. HAC standard errors are mandatory here: overlapping realised windows induce strong serial correlation in the residuals.

Kupiec.
Is the value-at-risk exception *count* right (Kupiec, 1995)?

Christoffersen.
Are the exceptions *independent*, or do they cluster (Christoffersen, 1998)?

Kupiec and Christoffersen are both reported because they fail differently, and the second failure is the dangerous one. A model can produce exactly the right number of breaches over a decade and place them all in a single week — which is how a risk system misses a crisis while passing its annual review.

6.6 Reading a Mincer-Zarnowitz slope

A slope below 1 is *partly mechanical* when volatility moves faster than the measurement horizon, because realised volatility over a forward window blends regimes while the forecast refers to one. Table 10 measures the effect on simulated data using a *perfect* point-in-time forecast and a 21-day horizon.

Table 10. Mincer-Zarnowitz slope under a perfect forecast, by regime persistence.

True regime length	MZ slope
21 days	0.50
126 days	0.93
504 days	1.02

A slope under 1 on real data is therefore not by itself evidence that the model over-reacts. The application states this where the statistic is displayed, because a diagnostic that is routinely misread is a diagnostic that needs its caveat attached to it rather than filed in a document.

6.7 Daily and multi-day risk

Factors that close at different times — the euro and Japanese curves, a London ETF, Yahoo's FX fixes — put one day's common shock into two days' returns. A daily covariance cannot see the second half and should not: a day's return does not contain tomorrow's catch-up. The variance of an h-day return does. Every forecast therefore carries two volatilities: the daily one, from which value at risk is read, and the h-day one,

which adds the lead-lag term of Appendix A.6 truncated at L lags and, for a Dimson model, uses the collapsed beta. The bias statistic, a statement about h-day returns, is read against the second, and an h-day QLIKE scores it against the squared h-day return — the only outcome that can see cross-day covariance at all. On simulated factors where one carries 60% of the other's move a day late, the unadjusted forecast's bias statistic is 1.30 and the adjusted one's within 0.12 of 1.

6.8 Portfolio-level validation

Every other test here scores one security's total volatility, which cannot tell a model whose factor covariance is right from one whose specific risks happen to make up for a wrong one, and says nothing about the off-diagonal structure that is the whole of what a factor model adds over a list of volatilities. Portfolios can. An equal-weighted sector basket is under-forecast if its stocks' residuals co-move and the model treats them as independent; a within-sector long-short pair is over-forecast for the same reason; a cross-sector spread is wrong if the factor covariance is.

Table 11. Portfolio-level validation, 54 US large caps, daily forecasts scored from January 2017 (run of 2026-10-13). Share of portfolios whose 21-day bias statistic lies in $1 \pm \sqrt{2/n}$, with the median statistic in brackets. The factor covariance half-life is 42 days, 21 in the fast columns.

	no sector	own sector	+ lead-lag	no sector, fast	own sector, fast
Market, equal weight	0% (1.15)	0% (1.14)	0% (1.14)	0% (1.19)	0% (1.18)
Sector baskets	64% (1.09)	73% (1.02)	82% (1.07)	55% (1.11)	73% (1.06)
Within-sector pairs	73% (0.94)	100% (1.02)	91% (1.03)	73% (0.94)	100% (1.02)
Cross-sector spreads	27% (1.17)	82% (1.02)	82% (1.06)	27% (1.19)	82% (1.03)
Random 10-name baskets	70% (1.11)	90% (1.10)	90% (1.10)	55% (1.13)	70% (1.12)
Residual corr., within sector	+0.222	-0.010	-0.010	+0.222	-0.010
Residual corr., across sectors	+0.006	+0.011	+0.011	+0.006	+0.011
First residual PC, share	0.082	0.050	0.050	0.082	0.050
Daily loss minus own EWMA's	+0.071	+0.042	+0.042	+0.027	-0.003

Table 11 is the case for the sector factors. Without them the out-of-sample residuals of two stocks in one sector correlate 0.22, the model treats that as diversifiable, and the portfolios it matters to fail in the predicted directions: sector baskets under-forecast, within-sector pairs over-forecast, and only 27% of cross-sector spreads calibrated. With each stock's own sector factor the residual correlation is -0.01 and the three families are calibrated at 73%, 100% and 82%. With a 21-day covariance half-life the model with sector factors also beats each portfolio's own EWMA volatility day by day on average, which no configuration without them does; the 42-day half-life sizes the 21-day risk and the 99% tail better. The market portfolio's 21-day risk is understated in every configuration while its daily risk is not.

The Risk Lens runs the same forecast on any portfolio of stocks and funds, each fund on its asset-class template. Five fund portfolios repeat the trade. At a 42-day half-life every one loses to its own EWMA volatility on daily QLIKE ($p \leq 0.005$). At 21 days four tie with it and the 60/40 still loses ($p = 0.027$). The fitted-t 99% VaR fails Kupiec for four of the five at either speed. Two of the five have bias statistics of 1.69 and 1.82, but almost all of that comes from the single 21-day block that began on 21 February 2020. Without that block they are 0.91 and 0.81. The view reports the worst block beside the statistic.

6.9 Reliability

With forty factors on a 252-day window, a period in which only a handful of factors yet exist leaves the design near-singular. Betas then explode in offsetting pairs, and $\beta'\Sigma\beta$ with them: one security produced a 360% predicted volatility this way, which then dominated the Mincer-Zarnowitz regression as a single leverage point.

Such forecasts are stored, flagged with the reason, and excluded from scoring — not silently dropped, and not silently used. The gate is a condition number above 200 on either panel, on standardised regressors, plus a maximum VIF above 100 on the orthogonalised panel only, for the reason given in Section 4.6. On the current panel it excludes no orthogonalised window and under 2% of raw ones.

7 How the terminal works

Everything described so far is a method. This section is the operating manual: what runs, when, in what order, and what a person has to do to keep the model current.

7.1 Two paths, and why they are separate

Work in this system divides cleanly into two kinds, and conflating them is the mistake the design exists to avoid.

The scheduled path refreshes data. It fetches prices and levels, rebuilds the forty factors on both bases, and re-runs the stationarity battery. It touches every series in the panel, it is the same work every night, and it takes about two minutes.

The on-demand path estimates a model. It regresses one security on the factor panel under one specification, builds the matching covariance, forecasts its risk and scores the forecast. It is per-security and per-specification, it is minutes of computation for a wide window rolled weekly, and there are more specifications than anyone would want recomputed nightly.

Why the nightly chain does not re-estimate loadings

A specification hashes to a `spec_id`, and a loading view an analyst has already seen is a table read rather than several hundred regressions. Re-estimating every stored specification each night would turn a two-minute refresh into an open-ended one, to recompute results nobody had asked for. Estimation is therefore triggered by the request that needs it, and the cache makes the second request free.

7.2 The pipeline

Figure 1 is the whole system. Read it top to bottom: vendors at the top, the two computation paths in the middle, the interface at the bottom.

Three properties of that diagram are load-bearing.

The chain is ordered and fails closed. Each stage is a prerequisite for the ones after it, and the run stops at the first fatal failure rather than continuing on half-refreshed inputs. A factor built on a partial refresh is worse than yesterday's factor: it looks current and is not. Only the diagnostics stage is advisory — a warning on one factor should not stop a nightly run, and the estimator reads the verdicts itself.

The store is the only shared state. Nothing in the on-demand path calls anything in the scheduled path, and neither calls the interface. A refresh that fails leaves yesterday's panel intact and every cached specification still readable.

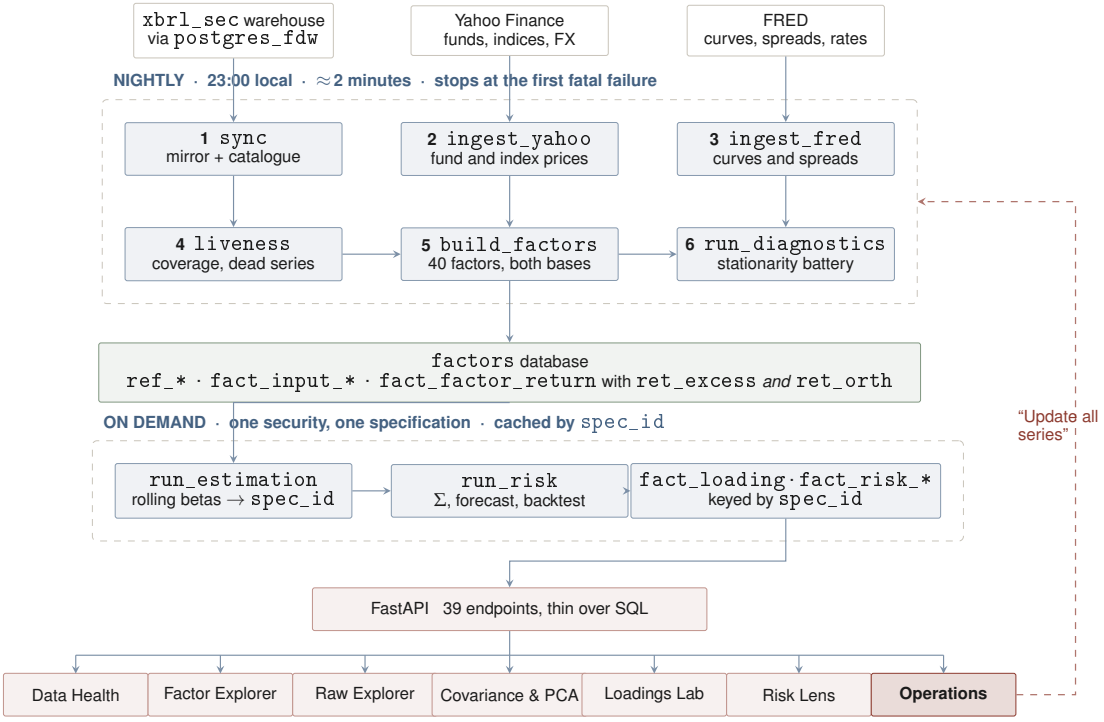


Figure 1. The terminal, end to end. Solid arrows are data flowing forward; the dashed arrow is the Operations tab starting the nightly chain on request. The store in the middle is the only state the two computation paths share, and it holds every factor twice — raw and orthogonalised — which is what lets a specification name its own panel and be scored against the matching covariance.

Both bases are stored, not derived. `fact_factor_return` holds `ret_excess` and `ret_orth` side by side. That is what allows a specification to declare which panel it was estimated on and be scored against the matching covariance (Section 4.4).

7.3 Updating every series, daily

There are three ways to run the chain, and they run the same code.

From the terminal's Operations tab. A single button, *Update all series*. The six stages appear as a checklist that fills in as they complete, the scheduler's own log streams underneath, and the page refuses to start a second run while one is in flight. The tab also answers the question that comes before the button: for each layer of the model — instrument returns, level series, factor returns — it shows the newest observation and how many days behind the panel it is, so that "does this need refreshing?" is visible rather than guessed.

From the command line. Identical work, identical exit code, and the exit code is meaningful — non-zero means a fatal stage failed, which is what makes it safe to put behind a system scheduler.

```
python -m backend.pipeline.scheduler --once
```

As a daemon. Runs the chain at 23:00 local: after the US close and the FRED daily refresh, and before midnight so the run is stamped with the day whose data it fetched.

```
python -m backend.pipeline.scheduler
```

Table 12. The nightly chain, with times from a representative run.

	Stage	What it does	Time
1	`sync`	Mirrors warehouse prices over the foreign-data wrapper and rebuilds the 9,434-row security catalogue	16s
2	`ingest_yahoo`	Total-return fund and index prices for every instrument a factor is built from	20s
3	`ingest_fred`	Treasury curves, the ICE BofA spread ladder, policy rates, financial-conditions indices	16s
4	`liveness`	Recomputes each instrument's last observation, so the gate sees what the two ingests just fetched	17s
5	`build_factors`	Constructs all forty factors on both bases	14s
6	`run_diagnostics`	Stationarity battery over factors and instruments *(advisory)*	40s
		Total	**≈125s**

A *full rebuild* — also a button, also a flag (`-full`) — ignores the watermarks and re-fetches every history from the start. It is minutes rather than seconds and is needed only after a schema change or a vendor restatement, which is why it is the secondary control rather than the primary one.

7.4 Estimating a security

The on-demand path is reached from the Loadings Lab. Choosing a security and pressing estimate does this:

1. The parameters on the page hash to a `spec_id`. If that identifier already has stored loadings, the response is a table read and the rest of this list does not happen.
2. If the security has never been priced here, it is pulled from the warehouse on the spot — one indexed read — and converted to the base currency.
3. The rolling regression runs: one fit per roll-forward step, each with its own diagnostics, stored against the `spec_id`.
4. Requesting risk for that specification builds the covariance *from the panel the specification names*, forecasts, scores the forecast against the forward outcome, and flags any window the reliability gate rejects.

The catalogue behind step 2 is why the search box offers 9,434 securities rather than the 203 the factor panel uses. It is materialised nightly by stage 1 because deriving it live means aggregating two price tables of fifteen million rows apiece — around eight seconds per keystroke.

7.5 What a refresh does not do

Stated so that a green run is not read as more than it is. The chain refreshes inputs and factors. It does not re-estimate loadings, does not re-score risk, and does not back-fill a history a vendor has restated. The first two are on-demand by design; the third is what the full rebuild is for. The Operations tab says all of this on the page, next to the button, rather than in a document the person pressing it has not read.

8 Software architecture

8.1 The layering, and why it is the main design decision

Layer	Contents	Constraint
`backend/core`	Statistics: 7 modules, 2,309 lines	Pure functions over arrays; no I/O
`backend/pipeline`	Ingestion, construction, estimation	May touch the database
`backend/app`	FastAPI routers, assistant	Thin; no statistics of its own
`sql`	15 idempotent migrations	Applied in order, fail-fast
`frontend`	Next.js 14, seven pages	No computation beyond presentation

The `backend/core` boundary is the main architectural decision in the project. Every statistical routine is a pure function over NumPy arrays with no database access, no configuration lookup and no logging side effects. That is what makes the accuracy claims in this paper checkable: a test can feed each routine a process whose truth is known analytically and assert that it reaches the right answer, without a database, a fixture or a network.

The alternative — statistics computed inside the query that fetches the data, which is the path of least resistance — would make every claim in Sections 4 and 6 unverifiable except by inspection.

8.2 Data model

The schema separates reference data (`ref_`), facts (`fact_`), staging (`stage_`), dimensions (`dim_`) and operational state (`etl_`). The factor panel lives in one table with both bases side by side:

```
fact_factor_return(factor_id, date, ret_excess, ret_orth, ...)
```

Storing both columns rather than deriving one on read is what allows a specification to name its panel and be scored consistently against it. Loadings, regression diagnostics, covariance matrices, risk forecasts and backtests are all keyed by `spec_id` (Section 5.4), so the entire estimation for one set of settings can be located, invalidated or recomputed as a unit.

Migrations are idempotent and applied in order by a fail-fast runner. The warehouse is reached through `postgres_fdw` rather than by copying: the security catalogue of 9,434 rows is materialised nightly, because deriving it live means aggregating two price tables of fifteen million rows apiece — around eight seconds per keystroke in a search box.

8.3 Pipeline jobs

`sync`	Mirror the warehouse over the foreign-data wrapper; pull securities on demand
`ingest_yahoo`, `ingest_fred`	Refresh the factor universe and the level series
`build_factors`	Construct all forty factors and both bases
`run_diagnostics`	Run the stationarity battery over factors and instruments
`run_estimation`	Rolling regression for one instrument under one specification
`run_risk`	Covariance, forecast, backtest and reliability for one specification
`scheduler`	The nightly sequence, as one command

Each job writes an entry to an operational run log with its mode, scope and per-item state, so a partial failure can be identified and resumed rather than re-run from the beginning. The application surfaces this: selecting a run in the interface shows what that run did, with its key parameters rendered as chips.

8.4 API

FastAPI over asyncpg, with the routers kept deliberately thin — they translate between HTTP and SQL and do no statistics. Thirty-nine endpoints across eight routers cover factor series and diagnostics, raw source series, the covariance matrix and its principal components, loadings estimation, risk, metadata, operations, and the assistant.

The estimation endpoint is the one long-running call: a wide window rolled weekly is upward of a thousand regressions, and the request does not return until every one is written. The interface reports elapsed seconds rather than a static spinner, because a motionless "Estimating…" for half a minute is indistinguishable from a hung page.

8.5 Interface

Seven pages — Data Health, Factor Explorer, Raw Explorer, Covariance & PCA, Loadings Lab, Risk Lens and Operations — on Next.js 14 with Plotly for charts. Section 7 describes what the last of them is for. Three conventions run through all the others.

Colour is a claim, not decoration. Colour on a diagnostic says whether the number is what the model wants, and the rule lives in a single module so that two pages cannot colour the same p-value differently. There are three tones and the third carries the weight: green where the test says what we want, amber where it flags something nothing is gated on, and plain where the outcome is expected. A rejected ARCH-LM or Jarque-Bera test is never red — volatility clustering and fat tails are the normal condition of a daily return series. Only the joint ADF $\times$ KPSS verdict, a wrong VaR breach count, and a design past the condition-number limit earn red.

Every metric explains itself. Each headline figure is a card that, on hover, states what the metric is, how it is computed and how to read it. The intent is that a number never appears without its definition being one gesture away.

Provenance is visible. The Raw Explorer shows the source system, the original vendor label and the full available history for every series, alongside the Python that turns it into a factor — extracted from the live source at request time, so it cannot fall out of date with the code that runs.

8.6 The assistant

An optional panel answers questions about the methodology and about the numbers currently on screen. It is grounded two ways: a methodology corpus assembled at startup from the module docstrings in `backend/core` and the registry note on every factor, and a snapshot of exactly what the open page has rendered.

It has no tools and no database access of its own, which is deliberate. The system prompt makes the supplied figures authoritative over the model's own arithmetic, so when asked for something not in view it names the page that produces it rather than estimating. Three live tests hold it to that contract.

8.7 Testing

The suite is 517 tests. They do not check that functions return numbers; each feeds a statistical routine a process whose truth is known and asserts that it reaches the right conclusion:

- a random walk must fail the stationarity gate; an AR(1) with $\phi = 0.5$ must pass
- a GARCH series must flag ARCH effects *without* failing
- a smoothed series must show a variance ratio above 1
- a forecast at half the true volatility must produce a bias statistic of 2
- clustered VaR breaches must be caught by Christoffersen even where Kupiec sees nothing wrong with the count
- shrinkage must beat the sample covariance when factors outnumber observations
- every rendered factor formula, evaluated independently, must reproduce the builder's output

This approach earned its keep. Five defects were found by tests or diagnostics rather than by inspection, and each would have silently corrupted results:

1. The variance-ratio implementation in the `arch` package consumes a *price level* and differences internally. Passing returns differenced twice, giving VR $\approx 1/q$ and flagging every clean factor as mean-reverting.
2. The ridge penalty was scaled by σ^{-2} instead of σ^2, so $\lambda = 1$ shrank every loading to zero.
3. Crypto weekend prints polluted the trading calendar, cutting the covariance matrix from 39 factors to 8.
4. `fx_carry` was the yen short wearing another name — correlation -0.87 with `fx_jpy` before it was residualised against it.
5. Stability columns held IEEE NaN rather than NULL, because pandas converts `None` to NaN in a float column and `double precision` accepts it. `IS NULL` never matched, `count()` counted them, and a single NaN turned an average over the whole column into NaN.

None of the five would have raised an error. Each produced a plausible number.

8.8 Operations and reproducibility

The nightly refresh is a single command. A launcher script starts the API and the web application, waits until both answer, and opens the browser; anything already serving is left alone. It carries one non-obvious check. A production Next.js server reads its build manifest once, at boot, so rebuilding the frontend while it runs leaves it serving HTML that points at the previous build's hashed chunks. Those files no longer exist, so the assets 404 — a missing stylesheet renders the page as unstyled HTML, a missing script renders nothing at all, and neither reports anything. Port-probing cannot detect it, because the server answers 200 perfectly well. The script therefore requests every route's HTML, follows every asset each one references, and restarts the web application when any of them fails to resolve; the route list is read from the prerendered pages on disk, so adding a page cannot leave it unchecked.

9 Limits

Style proxies are long-only funds.
Not long-short portfolios. Validated correlations after orthogonalisation: momentum 0.64, value 0.50, low volatility 0.20, quality 0.14. The last two are weak and should not be leaned on.

FX carry is an approximation.
Three policy-rate differentials in place of a G10 basket built from forward points, with a residual yen dependence of $+0.16$ after residualisation.

Euro-area and Japanese rates lag.
They are warehouse-sourced with no refresh in this project.

Short histories.
Style factors begin in 2011–2013 and `liq_funding` in 2018. Any window longer than the shortest factor's history silently drops it; the usable factor count is recorded per window for this reason.

Portfolio validation is one universe, one configuration.
Fifty-four US large caps on the equity template, with and without sector factors. The model agent's per-security choices are not yet aggregated.

The lead-lag lag count rests on few h-day returns.
A validation span holds 60 to 150 of them; the count stays at zero unless the evidence is clear.

Breach clustering is unresolved.
Both panels fail Christoffersen. Applying a 21-day-horizon forecast to daily returns produces dependent exceptions by construction, and the honest description is that the horizon and the test are mismatched rather than that the model passes.

No optimiser, no attribution, no macro state space.
See below.

The model can only measure risks that have appeared in returns.
Derivatives, concentration and tail positions remain invisible. This is why specific risk has a floor and why model uncertainty is surfaced rather than hidden.

Three components a factor model of this kind is often expected to carry are absent deliberately rather than by oversight.

Portfolio optimisation and performance attribution. The covariance matrix is optimiser-ready — positive semi-definite, with interpretable exposures — but no optimiser sits on top of it, and no attribution layer decomposes a realised return into factor contributions. Both are downstream consumers of the model rather than parts of it, and building either before the risk forecast has been shown to be right would be building on an unverified base.

A mixed-frequency state-space layer. Macro release surprises are implemented only as the sparse release-event transform of Section 2. There is no announcement-window lead-lag structure and no Kalman-filtered latent macro state interpolating releases onto a daily grid. Interpolation is exactly the forward fill this model refuses elsewhere, with a filter in front of it; doing it properly means committing to a state equation that the data available here cannot identify.

A Statistical methodology

This appendix states each procedure in the form in which it is implemented, including the conventions — bandwidth rules, degrees of freedom, boundary handling — that decide whether two nominally identical implementations agree. Every routine named here is a pure function over arrays in `backend/core` and is exercised by the test suite against a process whose answer is known analytically.

A.1 Conventions

The significance level is $\alpha = 0.05$ throughout, and every p-value is stored and displayed so that a reader may disagree with the threshold. Returns are natural logarithms. Annualisation uses 252 trading days: a daily standard deviation is multiplied by $\sqrt{252}$ and a daily mean by 252. Sample standard deviations use ddof $= 1$ unless stated; the Ledoit-Wolf derivation is the exception and is noted where it arises.

A.2 The stationarity battery

A single Dickey-Fuller test is useless as a gate on daily returns: it rejects almost always, so it cannot discriminate. The battery is therefore built around what actually goes wrong with financial series — an untransformed level, a structural break, stale pricing, illiquidity — with each test given the role its own null hypothesis supports.

A.2.1 Unit-root tests, read jointly

Augmented Dickey-Fuller (Dickey and Fuller, 1979).
H_0: a unit root is present. Lag order is selected by AIC; the regression includes a constant, which is the right specification for an excess return with a possible non-zero mean.

KPSS (Kwiatkowski et al., 1992).
H_0: the series *is* stationary — the exact complement of ADF. Level stationarity, with the automatic Newey-West bandwidth.

Phillips-Perron (Phillips and Perron, 1988).
Same null as ADF but with a non-parametric correction for serial correlation and heteroskedasticity in place of ADF's augmentation lags. Reported alongside as a robustness check, not gated on.

ADF and KPSS have opposite nulls, and reading them together yields four verdicts where either alone yields two. This is the single most useful idea in the battery:

Table 13. Joint reading of ADF and KPSS at $\alpha = 0.05$.

ADF	KPSS	Verdict	Consequence
rejects	does not reject	stationary	pass
does not reject	rejects	**unit root**	**fail** — must be differenced
rejects	rejects	break or heteroskedasticity	warn
does not reject	does not reject	inconclusive — usually too short	warn

Only the second row blocks a series. The third is the interesting one: both tests rejecting is not a contradiction but a signature — it indicates a structural break or strong het-

eroskedasticity rather than a clean unit root, and it is the only condition under which the Zivot-Andrews test is run.

A.2.2 Structural break

Zivot-Andrews (Zivot and Andrews, 1992) tests a unit root against stationarity around a single endogenously determined break, searching every candidate break date and taking the minimum statistic. It is by far the most expensive test in the battery, which is why it runs only when the joint reading says "not a clean unit root" and at least 100 observations are available. The estimated break date is stored, because a break in 2008-09 and a break in 2020-03 invite different conclusions.

A.2.3 Stale pricing

Lo-MacKinlay variance ratio (Lo and MacKinlay, 1988), heteroskedasticity-robust, at lags $q \in \{2, 5, 10\}$:

$$\mathrm{VR}(q) \;=\; \frac{\mathrm{Var}\left(\sum_{j=0}^{q-1} r_{t-j}\right)}{q\,\mathrm{Var}(r_t)}. \tag{11}$$

Under a random walk $\mathrm{VR} = 1$. Values above 1 indicate positive autocorrelation — trending or smoothed pricing — and values below 1 mean reversion. The heteroskedasticity-robust variant is essential: daily returns are conditionally heteroskedastic by construction, and the homoskedastic version rejects on that alone.

A level, not a return

The `arch` *package's variance-ratio implementation consumes a* price level *and differences internally. Passing returns to it differences twice, giving* $\mathrm{VR} \approx 1/q$ *and flagging every clean factor as mean-reverting. The implementation here therefore reconstructs a level by cumulative summation before calling it, and a test asserts that a smoothed series shows* $\mathrm{VR} > 1$ *— which is how the defect was found.*

Ljung-Box (Ljung and Box, 1978) at 10 lags, H_0: no autocorrelation up to lag 10. It is the second stale-pricing signal, and is also computed on squared returns, where rejection indicates volatility clustering rather than predictability in the mean.

A.2.4 Conditional heteroskedasticity

Engle's ARCH-LM (Engle, 1982) at 10 lags, H_0: no conditional heteroskedasticity. *This test is recorded and never gated on.* A GARCH process is strictly stationary; gating on ARCH-LM would exclude essentially every financial series ever published. The reason to know about clustering is that it argues for HAC standard errors — not against the factor.

A.2.5 Distribution and liquidity

Skewness, excess kurtosis (Fisher convention, zero for a Gaussian) and the Jarque-Bera statistic are recorded. Like ARCH-LM, Jarque-Bera is never gated on: fat tails are the normal condition of a daily return series.

The **zero-return share** is the cheapest and frequently the most telling check in the battery. Above 50% the series is not really trading daily and the verdict fails; above 20% it warns. The **maximum absolute return** is recorded as a data-integrity check — a single print of -5.775 in log space is not a market event but a corporate action or a bad first row.

A.2.6 Verdict assembly

Condition	Verdict	Flag
Joint reading is "unit root"	fail	`unit_root`
Fewer than 60 observations	fail	`too_short`
Zero variance	fail	`constant`
Zero-return share > 0.50	fail	`illiquid`
Joint reading inconclusive	warn	`inconclusive_unit_root`
Joint reading is "break"	warn	`structural_break`
Fewer than 252 observations	warn	`short_history`
Zero-return share > 0.20	warn	`sparse_trading`
$\lvert \mathrm{VR} - 1 \rvert > 0.25$ with a significant p	warn	`stale_pricing`

Every flag carries a human-readable reason, and the reason rather than the flag is what the interface displays.

A.2.7 Sparse release factors

A release-event factor is zero on roughly 80% of days by construction. The liquidity and stale-pricing gates would read that as a dead instrument. For a series marked sparse, the tests therefore run on the non-zero subsequence — the statistically meaningful object is the sequence of releases — and the sparsity is reported rather than penalised. A short window over such a factor holds few releases, so it warns instead of failing, with the reason directing the reader to the full-history diagnostic where power is adequate.

A release factor is dated by its *publication*, not by the period it describes. FRED files an NFCI reading under the Friday ending its reference week; the Chicago Fed publishes it the following Wednesday. The series is therefore read from ALFRED as first releases, each placed on the day it was published and rolled to the next trading day over a holiday; observations older than the first vintage are placed five days after their reference date. On the Friday dating the factor's changes correlated -0.33 with the five days of equity returns before publication; on the publication dating, -0.11.

A.2.8 Volatility surprise

A volatility factor without a traded series (`vol_rates`) is

$$f_t = \frac{|r_t|}{s_{t-1}} - \sqrt{2/\pi}, \qquad s_t^2 = \lambda s_{t-1}^2 + (1-\lambda)\, r_t^2, \quad \lambda = 0.5^{1/21}, \tag{12}$$

zero in expectation when the forecast is right. The daily change in a 21-day rolling volatility, which it replaced, moves by $(r_t^2 - r_{t-21}^2)/21$ to first order: half of it is the squared return leaving the window, an echo with autocorrelation -0.44 at lag 21.

A.3 Regression

A.3.1 Weighted least squares

With design matrix X (a column of ones followed by the factor panel) and weights w normalised to mean one, the estimator solves the weighted normal equations. Equal weighting sets $w_t = 1$; exponential weighting sets

$$w_t \propto 2^{-(T-t)/H}, \tag{13}$$

for half-life H in days, again normalised to mean one so that the effective sample size is comparable across settings.

A.3.2 Huber

Robust regression (Huber, 1964) by iteratively reweighted least squares, five iterations, with tuning constant $\delta = 1.345$ — the value giving 95% asymptotic efficiency under Gaussian errors. The weight function is

$$\psi(z) = \begin{cases} 1 & |z| \le \delta \\ \delta/|z| & |z| > \delta \end{cases}, \qquad z_t = \frac{\varepsilon_t}{\hat{s}}, \tag{14}$$

with $\hat{s}$ a robust scale estimate of the residuals. Observations inside δ robust standard deviations keep unit weight; those outside are downweighted in proportion to their distance rather than discarded.

A.3.3 Ridge

The penalty applies to standardised slopes so that λ does not depend on whether a factor is quoted in percent or in decimals, and the intercept is left unpenalised so that α is not biased toward zero. Penalising $\sum_j (b_j \sigma_j)^2$ gives normal equations

$$\left(X'X + \lambda n \,\mathrm{diag}(\sigma^2)\right) b = X'y, \tag{15}$$

so the penalty is *multiplied* by σ^2.

The sign of an exponent

The original implementation divided by σ^2 instead. For daily returns with $\sigma \approx 0.01$ that inflates the penalty by a factor of 10^4, and $\lambda = 1$ shrank every loading to exactly zero. The regression still returned, still reported an R^2, and still produced a loading vector; it was simply the zero vector. A test that asserts a modest ridge penalty leaves a known loading recognisable is what caught it.

A.3.4 HAC standard errors

Newey-West (Newey and West, 1987) with Bartlett (triangular) weights:

$$V = (X'X)^{-1}\left[S_0 + \sum_{\ell=1}^{L} w_\ell \left(\Gamma_\ell + \Gamma_\ell'\right)\right](X'X)^{-1}, \qquad w_\ell = 1 - \frac{\ell}{L+1}, \tag{16}$$

where Γ_ℓ is the lag-ℓ autocovariance of the score vectors $\varepsilon_t x_t$. Bartlett weights guarantee a positive semi-definite result, which a truncated (unweighted) kernel does not. The automatic bandwidth is

$$L = \left\lfloor 4\left(\frac{T}{100}\right)^{2/9} \right\rfloor, \tag{17}$$

giving 4 lags at 100 observations, 5 at 252 and 6 at 1000. The bandwidth is deliberately small: autocorrelation in daily returns is short-lived, and the cost of over-specifying L is a noisy covariance estimate.

A.3.5 Dimson lead-lag

The design is augmented with L lagged copies of the factor panel and the coefficients are summed factor by factor (Section 5.1). Only lags are included; a lead would use tomorrow's factor return to explain today's security return, which is unavailable at forecast time.

The standard error of the sum uses the full coefficient covariance:

$$\mathrm{se}\left(\hat\beta_j^{\mathrm{Dimson}}\right) = \sqrt{\textstyle\sum_\ell \sum_m \mathrm{Cov}\left(\hat\beta_j^{(\ell)}, \hat\beta_j^{(m)}\right)}. \tag{18}$$

The lag coefficients are typically negatively correlated, so the diagonal-only approximation $\sqrt{\sum_\ell \mathrm{se}^2}$ overstates the uncertainty. It is used only as a conservative fallback when no covariance is supplied, and the caller is told to treat the resulting interval as wide.

A.3.6 Collinearity diagnostics

The **condition number** is the ratio of the largest to the smallest singular value of the standardised design matrix. It is the invariant that governs how far $\beta'\Sigma\beta$ can be amplified by near-collinearity, and it is what the reliability gate of Section 6.9 reads.

Variance inflation factors are computed from the inverse correlation matrix,

$$\mathrm{VIF}_j = \left[R^{-1}\right]_{jj}, \qquad R = \mathrm{corr}(X), \tag{19}$$

which is algebraically identical to $1/(1 - R_j^2)$ from k auxiliary regressions but costs one matrix inversion rather than k least-squares fits — a material difference at $k = 40$ over a thousand rolling windows. The auxiliary-regression form is retained in the code as the reference implementation, and a test asserts the two agree.

A.3.7 Beta stability

Between consecutive windows, on the factors common to both:

$$d_{L_1} = \sum_{j \in C} \left| \beta_j^{(t)} - \beta_j^{(t-1)} \right|, \qquad \rho = \operatorname{corr}\left(\beta_C^{(t)}, \beta_C^{(t-1)} \right), \qquad |C| \text{ recorded.} \tag{20}$$

Recording $|C|$ is not optional: d_{L_1} falls mechanically as the common basis narrows, so the two must be read together.

A.4 Orthogonalisation

Given a target series y and a regressor block X drawn from the levels above it in the hierarchy, the rolling residual is computed as follows.

1. Coefficients are refitted at t whenever t has reached the next refit point, on the trailing window $[\max(0,\, t-504),\, t)$, requiring at least $\max(252,\, k+2)$ complete observations. The window is half-open: the fit uses data strictly before t.
2. The next refit point is set to $t + 21$.
3. The residual at t is $y_t - \hat{a} - \hat{b}' X_t$, using whichever coefficients are currently in force.
4. Before the first successful fit, the residual is undefined rather than equal to the raw series — a factor's orthogonalised history starts when it can be estimated, not before.

Two variants exist for comparison and are exercised by the test suite: an expanding window, and an exact full-sample fit. Table 5 reports what each costs. The full-sample variant is the only one that achieves exact orthogonality, and it is the only one that uses future information; it is never used to build the panel.

A Gram-Schmidt routine is also available, which orthogonalises a panel's columns sequentially left to right. It is used for diagnostics rather than construction, because the block hierarchy — not column order — is what encodes the economic argument about which factor should absorb shared variation.

A.5 Covariance estimation

A.5.1 Sample

The unbiased sample covariance on complete rows, annualised by 252. Included as a baseline and as the reference against which shrinkage is tested; it is not a sensible choice at $K = 39$ on a rolling window.

A.5.2 Exponentially weighted

RiskMetrics-style, with weights $w_t \propto 2^{-(T-t)/H}$ normalised to sum to one and a default half-life of $H = 60$ days. Two details matter.

Deviations are taken from the *weighted* mean rather than the simple mean, so the estimate is internally consistent. And the result is bias-corrected for weighted sampling,

$$\hat{\Sigma} = \frac{\sum_t w_t (x_t - \bar{x}_w)(x_t - \bar{x}_w)'}{1 - \sum_t w_t^2}, \tag{21}$$

since $1 / \sum_t w_t^2$ is the effective sample size when the weights sum to one.

Sixty days, not eleven

The classic RiskMetrics $\lambda = 0.94$ corresponds to a half-life of about 11 days. That is too jumpy for a risk model used to set limits: the forecast moves materially on a single day's return. Sixty days is the usual compromise and is the default here.

A.5.3 Ledoit-Wolf shrinkage

Shrinkage toward a *constant-correlation* target (Ledoit and Wolf, 2003). The target F keeps each factor's own variance and replaces every correlation by the average correlation $\bar{r}$:

$$F_{ij} = \bar{r}\, s_i s_j \quad (i \neq j), \qquad F_{ii} = s_i^2. \tag{22}$$

This is a considerably better description of a factor panel than the identity target used in the simpler version of the method, which asserts that the factors are uncorrelated — precisely the assumption a factor panel violates.

The optimal intensity is

$$\delta^\star = \max\left\{0,\ \min\left\{1,\ \frac{\pi - \rho}{\gamma} \cdot \frac{1}{T}\right\}\right\}, \qquad \hat{\Sigma} = \delta^\star F + (1 - \delta^\star) S, \tag{23}$$

where π is the summed asymptotic variance of the sample covariance entries, ρ the asymptotic covariance between the target's estimation error and the sample's, and $\gamma = \|F - S\|_F^2$ the squared misspecification of the target. The derivation assumes the maximum-likelihood scaling for S (divisor T), which is what the implementation uses internally; the result is converted to the unbiased scaling (divisor $T - 1$) before being returned, so that it is directly comparable with the sample estimator. The realised $\delta^\star$ is stored on the result — it is informative in itself, since a high intensity says the sample estimate was carrying little signal.

A.5.4 Blend

$\Sigma = a\,\Sigma_{\text{EWMA}} + (1 - a)\,\Sigma_{\text{shrunk}}$, default $a = 0.6$. The EWMA leg supplies responsiveness to the current regime; the shrunk leg supplies conditioning. This is the default estimator.

A.5.5 Positive semi-definiteness

Every estimate is checked. Where the smallest eigenvalue is negative, the matrix is repaired by clipping eigenvalues at a small positive floor and reconstructing, then *rescaled* so that the diagonal returns to the original factor variances.

The rescale is the part that is easy to omit and important to keep: clipping alone raises the diagonal, which raises every volatility computed from the matrix. The repair is recorded on the result rather than applied silently.

A.5.6 Specific risk

$$\sigma_\varepsilon^2 \;=\; \max\{\sigma_{\text{floor}}^2,\ \hat{\sigma}_\varepsilon^2\}\,, \tag{24}$$

with $\sigma_{\text{floor}} = 2\%$ annualised. $\hat{\sigma}_\varepsilon^2$ is the loading window's residual variance (degrees-of-freedom corrected) or $\text{EWMA}_H\left[e_t^2 - f_t' V f_t\right] \cdot 252$ over the out-of-sample residuals e_t of the betas in force, the correction removing the beta estimation noise each residual contains. A shrinkage toward the median of every specification's windows — past and future, over unrelated securities — was removed in October 2026. Both the raw and the floored values are returned, along with a flag recording whether the floor bound, so that a floored number is never mistaken for a measured one.

A.5.7 Lead-lag covariance

Markets closing at different times put one day's common shock into two days' returns. The variance of an h-day sum is exactly $h\left[\Gamma_0 + \sum_{l=1}^{h-1}(1 - l/h)(\Gamma_l + \Gamma_l')\right]$, $\Gamma_l = \mathrm{E}[(f_t - \mu)(f_{t-l} - \mu)']$. The h-day factor covariance, per day, is

$$\Sigma_F^{(h)} \;=\; \Sigma_F \;+\; \sum_{l=1}^{L}\left(1 - \tfrac{l}{h}\right)(\hat{\Gamma}_l + \hat{\Gamma}_l'), \tag{25}$$

with $\hat{\Gamma}_l$ weighted as the method weights its own sample (EWMA weights by the age of t, equal weights, or the blend's mixture), estimated only from pairs of complete adjacent days, and the sum repaired to positive semi-definite because truncation at L loses the guarantee of the full Bartlett sum. Σ_F alone remains the covariance of a *daily* return and VaR is read from it. For a Dimson design the h-day forecast uses the collapsed beta $\sum_l b_l$, whose cross-lag terms belong to a multi-day return. L is chosen by the model agent after its search, as the fewest lags whose validation h-day loss is within one standard error of the best.

A.5.8 Assembly and attribution

$$\Sigma_r = B\,\Sigma_F\,B' + \text{diag}(\sigma_\varepsilon^2), \qquad \sigma_p = \sqrt{\beta'\Sigma_F\beta + \sigma_\varepsilon^2}. \tag{26}$$

Risk contributions use the standard Euler decomposition, $\text{RC}_j = \beta_j\,(\Sigma_F\beta)_j/\sigma_p$, which sums to the factor part of total risk exactly. Contributions are aggregated to block level for display, because a forty-row contribution table is not a decomposition anyone reads.

A.6 Forecast evaluation

A.6.1 Realised volatility and alignment

For horizon h, realised volatility at t is the annualised standard deviation of $r_{t+1}, \ldots, r_{t+h}$. The forecast it is compared against uses loadings from a window ending at or before t and a covariance estimated on returns up to t. The forward window is constructed by an explicit shift rather than by a rolling operation with a default alignment, because an off-by-one in this direction is lookahead and produces a flatteringly good result.

A.6.2 Portfolio aggregation

For portfolio weights w over stocks with betas B_t (zero where a stock has no exposure), held from each stock's latest window ending at or before t,

$$\hat{\sigma}^2_{p,t+1} \;=\; w' B_t \, \Sigma_{F,t} \, B_t' w \;-\; \sum_i w_i^2 \, \mathrm{tr}\big(\Sigma_{F,t} V_{i,t}\big) \;+\; \sum_i w_i^2 \, \hat{\sigma}^2_{\varepsilon,i,t}, \tag{27}$$

with $\Sigma_{F,t}$ a causal EWMA over pairs of observed factors, so a factor that starts late enters each pair from its first common day. Each portfolio is scored like a security: the standard deviation of $r_{p,t+1}/\hat{\sigma}_{p,t}$, daily QLIKE against the portfolio's own EWMA volatility, Kupiec at 99% with a normal and with a Student t fitted every h days to the trailing 756 standardised returns, and the bias statistic on non-overlapping h-day returns. The out-of-sample residuals $e_{i,t+1} = r_{i,t+1} - b_{i,t}' f_{t+1}$ are summarised by their mean pairwise correlation within and across sectors and by the principal components of the standardised residual panel.

A.6.3 Bias statistics

Two are computed. The **mean bias ratio** is $\overline{\sigma^{\text{real}}/\sigma^{\text{pred}}}$; it is intuitive but noisy, being a ratio of volatilities and therefore right-skewed. The **bias statistic** is the sharper and is what the text quotes:

$$z_t = \frac{r_t}{\sigma_t^{\text{pred}}}, \qquad \text{bias} = \mathrm{sd}(z_t), \tag{28}$$

the standard deviation of returns divided by the volatility forecast in force at each date. It equals 1 when the model is calibrated and exceeds 1 when risk is underestimated. The kurtosis of z is recorded alongside it: a bias statistic of 1 with a kurtosis of 8 describes a model that is right on average and wrong in the tails.

A.6.4 Mincer-Zarnowitz

Realised variance regressed on predicted variance (Mincer and Zarnowitz, 1969):

$$\left(\sigma_t^{\text{real}}\right)^2 = a + b\left(\sigma_t^{\text{pred}}\right)^2 + u_t, \qquad H_0 : \; (a,b) = (0,1). \tag{29}$$

Run on variances rather than volatilities, because variance is what the model forecasts and what aggregates linearly. Individual t-tests for $a = 0$ and $b = 1$ are reported together with a joint Wald test, $W = (\hat{\theta} - \theta_0)' V^{-1} (\hat{\theta} - \theta_0) \sim \chi^2_2$.

HAC standard errors are not optional here. Overlapping realised windows — a 21-day forward window observed daily — induce strong serial correlation in u_t by construction, and ordinary standard errors would reject far too often. Section 6 gives the separate caveat on interpreting the slope.

A.6.5 Value at risk and expected shortfall

$$\mathrm{VaR}_t(\alpha) = -z_\alpha \frac{\sigma_t^{\mathrm{pred}}}{\sqrt{252}}, \tag{30}$$

under either a Gaussian quantile or a scaled Student-t quantile with 5 degrees of freedom, the latter being the more honest default for daily returns. Expected shortfall is computed at the 97.5% level under the same distributional choice.

A.6.6 Kupiec proportion of failures

(Kupiec, 1995) A likelihood ratio on the exception *count*, χ^2_1:

$$\mathrm{LR}_{\mathrm{POF}} = -2\left[(T-x)\log(1-p) + x\log p - (T-x)\log(1-\hat{\pi}) - x\log\hat{\pi}\right], \tag{31}$$

with $p = 1 - \alpha$, x exceptions in T observations and $\hat{\pi} = x/T$. At the boundaries $x = 0$ or $x = T$ the ratio is degenerate; the implementation falls back to an exact two-sided binomial probability rather than returning a spurious statistic.

A.6.7 Christoffersen conditional coverage

(Christoffersen, 1998) H_0: the correct rate *and* independence. Transitions are counted as n_{ij} (from state i to state j, breach = 1), and

$$\mathrm{LR}_{\mathrm{CC}} = \mathrm{LR}_{\mathrm{POF}} + \mathrm{LR}_{\mathrm{IND}} \sim \chi^2_2. \tag{32}$$

Where the transition counts cannot identify the independence term — no breach ever follows a non-breach, or one of the states is never visited — the implementation reports Kupiec alone on one degree of freedom rather than pretending to test something the data cannot support.

This is the test that matters most and is reported even when it fails, because clustered breaches are the dangerous failure: a model can produce exactly the right number of exceptions over a decade and place them all in one week.

A.7 Density estimation

The distribution panel exists to answer one question — how far from Gaussian is this series, and where — so its defaults are chosen to keep the tails visible rather than to produce a smooth picture.

Robust scale. The smaller of the sample standard deviation and IQR/1.349. The two coincide for a Gaussian sample; on a fat-tailed series the second is much smaller, and taking the minimum ties bandwidth and bin width to the bulk of the distribution, which is where resolution is wanted.

Bin count. Freedman-Diaconis (Freedman and Diaconis, 1981), $h = 2\,\mathrm{IQR}\,n^{-1/3}$, with the resulting count clamped to $[20, 220]$ for legibility. Scott's rule uses the standard deviation and would be inflated by exactly the outliers that create the problem.

Bandwidth. Silverman's rule (Silverman, 1986) on the robust scale, $h = 0.9\,s\,n^{-1/5}$. Using the plain standard deviation here over-smooths a fat-tailed series into a flat blob.

Overlays and range. A Gaussian fit and a Student-t fit are drawn over the empirical density; the comparison between them is the whole purpose of the panel. The drawing range is the 0.1st to 99.9th percentile, and the observations outside it are *counted and reported*, not silently dropped — a plot that hid its own tail would be worse than the naive one it replaces.

B Factor registry

The forty factors, their block, their hierarchy level and the factors each is residualised against. Level 0 factors are used raw. A factor's orthogonalisation targets are always factors at a lower level, so the hierarchy is acyclic by construction and the panel can be built in one pass.

The construction of each factor — the exact formula over named instruments — is given in the companion document *factor-formulas*, which is generated from this same registry.

Factor	Block	Level	Residualised against
eq_global	Equity Market	0	— *used raw*
eq_cyclical	Equity Market	1	eq_global
eq_em	Equity Market	1	eq_global
eq_europe	Equity Market	1	eq_global
eq_japan	Equity Market	1	eq_global
eq_size	Equity Market	1	eq_global
eq_us	Equity Market	1	eq_global
sty_lowvol	Equity Style	2	eq_global, eq_us, eq_cyclical
sty_momentum	Equity Style	2	eq_global, eq_us, eq_cyclical
sty_quality	Equity Style	2	eq_global, eq_us, eq_cyclical
sty_value	Equity Style	2	eq_global, eq_us, eq_cyclical
rt_breakeven	Rates	0	rt_us_level
rt_ea_level	Rates	0	rt_us_level
rt_jp_level	Rates	0	rt_us_level
rt_uk	Rates	0	rt_us_level
rt_us_curve	Rates	0	— *used raw*
rt_us_level	Rates	0	— *used raw*
rt_us_slope	Rates	0	— *used raw*
cr_em_hard	Credit	1	rt_us_level, rt_us_slope, cr_hy, eq_global
cr_hy	Credit	1	rt_us_level, rt_us_slope, eq_global
cr_ig	Credit	1	rt_us_level, rt_us_slope
cr_loans	Credit	1	rt_us_level, cr_hy
cr_em_local	Credit	2	rt_us_level, cr_em_hard, fx_usd
fx_usd	FX	0	— *used raw*
fx_carry	FX	1	fx_usd, fx_jpy
fx_em	FX	1	fx_usd
fx_jpy	FX	1	fx_usd
cm_broad	Commodities	0	— *used raw*
cm_agri	Commodities	1	cm_broad
cm_energy	Commodities	1	cm_broad
cm_gold	Commodities	1	cm_broad
cm_industrial	Commodities	1	cm_broad
vol_equity	Volatility	1	eq_global
vol_rates	Volatility	1	rt_us_level

Factor	Block	Level	Residualised against
vol_variance_premium	Volatility	1	eq_global
liq_conditions	Liquidity & Stress	0	— *used raw*
liq_funding	Liquidity & Stress	0	— *used raw*
liq_risk_off	Liquidity & Stress	2	eq_global, rt_us_level, cr_hy
arp_trend	Alternative Risk Premia	3	eq_global, rt_us_level, cm_broad, fx_usd
arp_xs_momentum	Alternative Risk Premia	3	eq_global, arp_trend
sec_communication	Equity Sector	2	eq_global, eq_us, eq_cyclical
sec_discretionary	Equity Sector	2	eq_global, eq_us, eq_cyclical
sec_energy	Equity Sector	2	eq_global, eq_us, eq_cyclical
sec_financials	Equity Sector	2	eq_global, eq_us, eq_cyclical
sec_health	Equity Sector	2	eq_global, eq_us, eq_cyclical
sec_industrials	Equity Sector	2	eq_global, eq_us, eq_cyclical
sec_materials	Equity Sector	2	eq_global, eq_us, eq_cyclical
sec_real_estate	Equity Sector	2	eq_global, eq_us, eq_cyclical
sec_staples	Equity Sector	2	eq_global, eq_us, eq_cyclical
sec_tech	Equity Sector	2	eq_global, eq_us, eq_cyclical
sec_utilities	Equity Sector	2	eq_global, eq_us, eq_cyclical

C Factor construction formulas

Every factor in the model, written out: the arithmetic that builds it from named instruments, and the equation that residualises it against the blocks above it. Section 2 gives the reasoning behind the constructions; this states them.

This appendix is generated from the factor registry rather than transcribed, so it cannot drift from the code that builds the panel. The transcription from the builders is itself checked: each rendered formula is evaluated independently by the test suite and asserted to reproduce the builder's own output on simulated inputs, which is what makes an equation here a statement about the implementation rather than about an intention.

C.1 Reading these

Each factor is stored twice, and the two are different series. The raw column `fact_factor_return.ret_excess` holds f_t, the construction formula's own output. The orthogonalised column `fact_factor_return.ret_orth` holds $\tilde{f}_t$, after the block hierarchy has residualised it. The model may be estimated on either, and the choice is recorded on the spec and carried through to the covariance, because $\beta'\Sigma\beta$ requires Σ to be the covariance of the same series the βs refer to. A factor whose orthogonalisation list is empty has $\tilde{f}_t = f_t$ identically.

Symbol	Meaning
r_t^X	log total return of instrument X, on a price series already adjusted for dividends and splits
c_t	daily cash rate: `FRED:DFF` at t, annualised percent, divided by 100 and by 252
x_t^X	excess return over cash, $x_t^X = r_t^X - c_{t-1}$
Δ	first difference, $\Delta s_t = s_t - s_{t-1}$
f_t	the raw factor
$\tilde{f}_t$	the orthogonalised factor
τ	the most recent orthogonalisation refit at or before t

Every lag is deliberate. A quantity dated $t-1$ is there because using its value at t would put information into a return that was not available when the return was earned.

C.2 Equity Market

C.2.1 `eq_global` — Global Equity

Hierarchy level 0. Method `single`.

> MSCI ACWI total return less cash. The single global market factor.

Construction

$$f_t = x_t^{\text{ACWI}} \tag{33}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. ACWI is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{ACWI}	`ACWI`	instrument return: log total return of ACWI
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t \tag{34}$$

Nothing sits above this factor in the hierarchy, so the raw and orthogonalised panels hold the same series.

C.2.2 `eq_us` — US Equity (ex-global)

Hierarchy level 1. Method `single`.

Construction

$$f_t = x_t^{\text{IVV}} \tag{35}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. IVV is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{IVV}	`IVV`	instrument return: log total return of IVV
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\, \tilde{f}_t^{\text{eq_global}} \tag{36}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j \tilde{f}_s^{g_j} \right)^2 \tag{37}$$

with g_1 = `eq_global`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.2.3 `eq_europe` — Europe Equity (ex-global)

Hierarchy level 1. Method `single`.

Construction

$$f_t = x_t^{\text{VGK}} \tag{38}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. VGK is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{VGK}	`VGK`	instrument return: log total return of VGK
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\, \tilde{f}_t^{\text{eq_global}} \tag{39}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j \tilde{f}_s^{g_j} \right)^2 \tag{40}$$

with g_1 = `eq_global`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.2.4 `eq_japan` — Japan Equity (ex-global)

Hierarchy level 1. Method `single`.

Construction

$$f_t = x_t^{\text{EWJ}} \tag{41}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. EWJ is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{EWJ}	`EWJ`	instrument return: log total return of EWJ
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\, \tilde{f}_t^{\text{eq_global}} \tag{42}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j \tilde{f}_s^{g_j} \right)^2 \tag{43}$$

with g_1 = `eq_global`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.2.5 `eq_em` — EM Equity (ex-global)

Hierarchy level 1. Method `single`.

Construction

$$f_t = x_t^{\text{EEM}} \tag{44}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. EEM is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{EEM}	`EEM`	instrument return: log total return of EEM
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\, \tilde{f}_t^{\text{eq_global}} \tag{45}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j \tilde{f}_s^{g_j} \right)^2 \tag{46}$$

with g_1 = `eq_global`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.2.6 `eq_size` — Size (small minus large)

Hierarchy level 1. Method `spread`.

Russell 2000 minus Russell 1000. Self-financing, so no cash leg.

Construction

$$f_t = r_t^{\text{IWM}} - r_t^{\text{IWB}} \tag{47}$$

1. Self-financing, so no cash leg: the funding cost cancels between the two legs and subtracting it would understate the spread.

Symbol	Source	Meaning
r_t^{IWM}	`IWM`	long leg: log total return of IWM
r_t^{IWB}	`IWB`	short leg: log total return of IWB

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\, \tilde{f}_t^{\text{eq_global}} \tag{48}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \Big(f_s - a - \sum_j b_j\, \tilde{f}_s^{g_j} \Big)^2 \tag{49}$$

with g_1 = `eq_global`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.2.7 `eq_cyclical` — Cyclical minus Defensive

Hierarchy level 1. Method `basket`.

> Sector breadth without spending eleven factors on it.

Construction

$$f_t = \frac{r_t^{\text{XLI}} + r_t^{\text{XLB}} + r_t^{\text{XLE}} + r_t^{\text{XLF}} + r_t^{\text{XLY}}}{5} - \frac{r_t^{\text{XLP}} + r_t^{\text{XLU}} + r_t^{\text{XLV}}}{3} \tag{50}$$

1. Equally weighted long basket of 5: each leg carries $1/5$ of the notional.
2. Minus an equally weighted short basket of 3. Both sides carry the market direction, so it cancels and what survives is the spread between them.
3. Self-financing, so no cash leg.

Symbol	Source	Meaning
r_t^{XLI}	`XLI`	long leg: log total return of XLI
r_t^{XLB}	`XLB`	long leg: log total return of XLB
r_t^{XLE}	`XLE`	long leg: log total return of XLE
r_t^{XLF}	`XLF`	long leg: log total return of XLF
r_t^{XLY}	`XLY`	long leg: log total return of XLY
r_t^{XLP}	`XLP`	short leg: log total return of XLP
r_t^{XLU}	`XLU`	short leg: log total return of XLU
r_t^{XLV}	`XLV`	short leg: log total return of XLV

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\, \tilde{f}_t^{\text{eq_global}} \tag{51}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \Big(f_s - a - \sum_j b_j\, \tilde{f}_s^{g_j} \Big)^2 \tag{52}$$

with g_1 = `eq_global`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.3 Equity Style

C.3.1 `sty_value` — Value

Hierarchy level 2. Method `single`.

> Long-only ETF residualised to market and sector. An approximation to a true long-short HML; validated against F-F HML over the common sample.

Construction

$$f_t = x_t^{\text{VLUE}} \tag{53}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. VLUE is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{VLUE}	`VLUE`	instrument return: log total return of VLUE
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\,\tilde{f}_t^{\text{eq_global}} - \hat{b}_{2,\tau}\,\tilde{f}_t^{\text{eq_us}} - \hat{b}_{3,\tau}\,\tilde{f}_t^{\text{eq_cyclical}} \tag{54}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j\, \tilde{f}_s^{g_j} \right)^2 \tag{55}$$

with $g_1, \ldots, g_3$ = `eq_global`, `eq_us`, `eq_cyclical`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.3.2 `sty_momentum` — Momentum

Hierarchy level 2. Method `single`.

Construction

$$f_t = x_t^{\text{MTUM}} \tag{56}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. MTUM is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{MTUM}	`MTUM`	instrument return: log total return of MTUM
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\,\tilde{f}_t^{\text{eq_global}} - \hat{b}_{2,\tau}\,\tilde{f}_t^{\text{eq_us}} - \hat{b}_{3,\tau}\,\tilde{f}_t^{\text{eq_cyclical}} \tag{57}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j\, \tilde{f}_s^{g_j} \right)^2 \tag{58}$$

with $g_1, \ldots, g_3$ = `eq_global`, `eq_us`, `eq_cyclical`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.3.3 `sty_quality` — Quality

Hierarchy level 2. Method `single`.

Construction

$$f_t = x_t^{\text{QUAL}} \tag{59}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. QUAL is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{QUAL}	`QUAL`	instrument return: log total return of QUAL
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\,\tilde{f}_t^{\text{eq_global}} - \hat{b}_{2,\tau}\,\tilde{f}_t^{\text{eq_us}} - \hat{b}_{3,\tau}\,\tilde{f}_t^{\text{eq_cyclical}} \tag{60}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j\, \tilde{f}_s^{g_j} \right)^2 \tag{61}$$

with $g_1, \ldots, g_3$ = `eq_global`, `eq_us`, `eq_cyclical`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.3.4 `sty_lowvol` — Low Volatility

Hierarchy level 2. Method `single`.

Construction

$$f_t = x_t^{\text{USMV}} \tag{62}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. USMV is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{USMV}	`USMV`	instrument return: log total return of USMV
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\,\tilde{f}_t^{\text{eq_global}} - \hat{b}_{2,\tau}\,\tilde{f}_t^{\text{eq_us}} - \hat{b}_{3,\tau}\,\tilde{f}_t^{\text{eq_cyclical}} \tag{63}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \Big(f_s - a - \sum_j b_j\, \tilde{f}_s^{g_j} \Big)^2 \tag{64}$$

with $g_1, \ldots, g_3$ = `eq_global`, `eq_us`, `eq_cyclical`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.4 Rates

C.4.1 `rt_us_level` — US Rates Level

Hierarchy level 0. Method `curve`.

Equal unit-duration long across the curve, scaled to 5y duration.

Construction

$$f_t = 5 \cdot \frac{1}{4} \sum_T u_t^{(T)}, \qquad u_t^{(T)} = \frac{b_t^{(T)}}{D_{t-1}^{(T)}} \tag{65}$$

$$b_t^{(T)} = -D_{t-1}^{(T)}\,\Delta y_t^{(T)} + \tfrac{1}{2} C_{t-1}^{(T)} \left(\Delta y_t^{(T)}\right)^2 + \frac{y_{t-1}^{(T)} - c_{t-1}}{252} \tag{66}$$

1. Convert each tenor's yield change into a bond return, with duration and convexity evaluated at yesterday's yield. Section 2.2 requires every factor to be a return, and a yield change is not one.

2. Divide each leg by its own modified duration. Five basis points on a 30y is not the same trade as five on a 2y, and without this the long end would dominate every shape.
3. Level: an equal unit-duration long at every tenor (2y, 5y, 10y, 30y).
4. Scale back by 5 years of duration.

Symbol	Source	Meaning
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$
FRED:DGS2	`FRED:DGS2`	US_TSY par yield at 2y
FRED:DGS5	`FRED:DGS5`	US_TSY par yield at 5y
FRED:DGS10	`FRED:DGS10`	US_TSY par yield at 10y
FRED:DGS30	`FRED:DGS30`	US_TSY par yield at 30y

Orthogonalisation

$$\tilde{f}_t = f_t \tag{67}$$

Nothing sits above this factor in the hierarchy, so the raw and orthogonalised panels hold the same series.

C.4.2 `rt_us_slope` — US Rates Slope (10s-2s)

Hierarchy level 0. Method `curve`.

Duration-neutral steepener: long 10y, short 2y, equal duration.

Construction

$$f_t = 5 \cdot \left(u_t^{(10y)} - u_t^{(2y)}\right), \qquad u_t^{(T)} = \frac{b_t^{(T)}}{D_{t-1}^{(T)}} \tag{68}$$

$$b_t^{(T)} = -D_{t-1}^{(T)}\,\Delta y_t^{(T)} + \tfrac{1}{2} C_{t-1}^{(T)} \left(\Delta y_t^{(T)}\right)^2 + \frac{y_{t-1}^{(T)} - c_{t-1}}{252} \tag{69}$$

1. Convert each tenor's yield change into a bond return, with duration and convexity evaluated at yesterday's yield. Section 2.2 requires every factor to be a return, and a yield change is not one.
2. Divide each leg by its own modified duration. Five basis points on a 30y is not the same trade as five on a 2y, and without this the long end would dominate every shape.
3. Slope: long 10y against short 2y in equal units of duration, so a parallel shift nets to zero and only the steepening survives.

4. Scale back by 5 years of duration.

Symbol	Source	Meaning
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$
FRED:DGS2	`FRED:DGS2`	US_TSY par yield at 2y
FRED:DGS10	`FRED:DGS10`	US_TSY par yield at 10y

Orthogonalisation

$$\tilde{f}_t = f_t \tag{70}$$

Nothing sits above this factor in the hierarchy, so the raw and orthogonalised panels hold the same series.

C.4.3 `rt_us_curve` — US Rates Curvature (butterfly)

Hierarchy level 0. Method `curve`.

> Long the belly, short the wings, duration-neutral.

Construction

$$f_t = 5 \cdot \left(2u_t^{(5y)} - u_t^{(2y)} - u_t^{(30y)}\right), \qquad u_t^{(T)} = \frac{b_t^{(T)}}{D_{t-1}^{(T)}} \tag{71}$$

$$b_t^{(T)} = -D_{t-1}^{(T)}\,\Delta y_t^{(T)} + \tfrac{1}{2}C_{t-1}^{(T)}\left(\Delta y_t^{(T)}\right)^2 + \frac{y_{t-1}^{(T)} - c_{t-1}}{252} \tag{72}$$

1. Convert each tenor's yield change into a bond return, with duration and convexity evaluated at yesterday's yield. Section 2.2 requires every factor to be a return, and a yield change is not one.
2. Divide each leg by its own modified duration. Five basis points on a 30y is not the same trade as five on a 2y, and without this the long end would dominate every shape.
3. Curvature: long two units of the 5y belly against one unit each of the 2y and 30y wings. Neutral to both a parallel shift and a steepening.
4. Scale back by 5 years of duration.

Symbol	Source	Meaning
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$
FRED:DGS2	`FRED:DGS2`	US_TSY par yield at 2y
FRED:DGS5	`FRED:DGS5`	US_TSY par yield at 5y
FRED:DGS30	`FRED:DGS30`	US_TSY par yield at 30y

Orthogonalisation

$$\tilde{f}_t = f_t \tag{73}$$

Nothing sits above this factor in the hierarchy, so the raw and orthogonalised panels hold the same series.

C.4.4 `rt_ea_level` — EA Rates Level

Hierarchy level 0. Method `curve`.

ECB AAA curve. Warehouse-sourced, so it lags the US block.

Construction

$$f_t = 5 \cdot \frac{1}{4} \sum_T u_t^{(T)}, \qquad u_t^{(T)} = \frac{b_t^{(T)}}{D_{t-1}^{(T)}} \tag{74}$$

$$b_t^{(T)} = -D_{t-1}^{(T)} \, \Delta y_t^{(T)} + \tfrac{1}{2} C_{t-1}^{(T)} \left(\Delta y_t^{(T)}\right)^2 + \frac{y_{t-1}^{(T)} - c_{t-1}}{252} \tag{75}$$

1. Convert each tenor's yield change into a bond return, with duration and convexity evaluated at yesterday's yield. Section 2.2 requires every factor to be a return, and a yield change is not one.
2. Divide each leg by its own modified duration. Five basis points on a 30y is not the same trade as five on a 2y, and without this the long end would dominate every shape.
3. Level: an equal unit-duration long at every tenor (2y, 5y, 10y, 30y).
4. Scale back by 5 years of duration.

Symbol	Source	Meaning
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$
ECB:BUND_2Y	`ECB:BUND_2Y`	EA_AAA par yield at 2y
ECB:BUND_5Y	`ECB:BUND_5Y`	EA_AAA par yield at 5y
ECB:BUND_10Y	`ECB:BUND_10Y`	EA_AAA par yield at 10y
ECB:BUND_30Y	`ECB:BUND_30Y`	EA_AAA par yield at 30y

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau} \, \tilde{f}_t^{\text{rt_us_level}} \tag{76}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j \tilde{f}_s^{g_j} \right)^2 \tag{77}$$

with g_1 = `rt_us_level`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.4.5 `rt_jp_level` — JP Rates Level

Hierarchy level 0. Method `curve`.

MOF/BOJ curve. Warehouse-sourced, so it lags the US block.

Construction

$$f_t = 5 \cdot \frac{1}{4} \sum_T u_t^{(T)}, \qquad u_t^{(T)} = \frac{b_t^{(T)}}{D_{t-1}^{(T)}} \tag{78}$$

$$b_t^{(T)} = -D_{t-1}^{(T)} \Delta y_t^{(T)} + \tfrac{1}{2} C_{t-1}^{(T)} \left(\Delta y_t^{(T)} \right)^2 + \frac{y_{t-1}^{(T)} - c_{t-1}}{252} \tag{79}$$

1. Convert each tenor's yield change into a bond return, with duration and convexity evaluated at yesterday's yield. Section 2.2 requires every factor to be a return, and a yield change is not one.
2. Divide each leg by its own modified duration. Five basis points on a 30y is not the same trade as five on a 2y, and without this the long end would dominate every shape.
3. Level: an equal unit-duration long at every tenor (2y, 5y, 10y, 30y).
4. Scale back by 5 years of duration.

Symbol	Source	Meaning
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$
MOF_JP:JGB_2Y	`MOF_JP:JGB_2Y`	JP_JGB par yield at 2y
MOF_JP:JGB_5Y	`MOF_JP:JGB_5Y`	JP_JGB par yield at 5y
MOF_JP:JGB_10Y	`MOF_JP:JGB_10Y`	JP_JGB par yield at 10y
MOF_JP:JGB_30Y	`MOF_JP:JGB_30Y`	JP_JGB par yield at 30y

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau} \tilde{f}_t^{\text{rt_us_level}} \tag{80}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j \tilde{f}_s^{g_j} \right)^2 \tag{81}$$

with g_1 = `rt_us_level`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.4.6 `rt_uk` — UK Gilts

Hierarchy level 0. Method `single`.

> No daily gilt curve exists anywhere in the warehouse, so this is an ETF total return converted to base currency rather than a curve factor.

Construction

$$f_t = \left(x_t^{\text{IGLT.L}} + \Delta \ln e_t^{\text{GBP}} \right) \tag{82}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. IGLT.L is a funded long position, so its return has to be measured over cash.
2. Convert to the base currency by adding the log change in GBP/USD. In logs the conversion is a sum, with no cross term.

Symbol	Source	Meaning
$r_t^{\text{IGLT.L}}$	`IGLT.L`	instrument return: log total return of IGLT.L
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$
e_t^{GBP}	`GBP`	USD per unit of GBP; its log change converts the local return into the base currency, which is exact in logs

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau} \tilde{f}_t^{\text{rt_us_level}} \tag{83}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j \tilde{f}_s^{g_j} \right)^2 \tag{84}$$

with g_1 = `rt_us_level`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.4.7 `rt_breakeven` — Inflation Breakeven

Hierarchy level 0. Method `synthetic_bond`.

> 10y breakeven change, duration-scaled. Carry excluded: a breakeven is a spread between two yields and has no coupon of its own.

Construction

$$f_t = -D_{t-1}\,\Delta y_t + \tfrac{1}{2}C_{t-1}\,(\Delta y_t)^2 \tag{85}$$

1. Treat the series as the yield of a 10y par bond and price a day's move on it, with duration and convexity taken at $t-1$ so nothing about today's move leaks into its own pricing.
2. No carry leg. A breakeven is the difference between two yields and earns no coupon of its own, so the builder adds the carry term and subtracts it again; the two cancel exactly and the factor is a pure duration-scaled change.

Symbol	Source	Meaning
FRED:T10YIE	`FRED:T10YIE`	quoted yield series, in percent; y_t is that divided by 100

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\,\tilde{f}_t^{\text{rt_us_level}} \tag{86}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \Big(f_s - a - \sum_j b_j\,\tilde{f}_s^{g_j}\Big)^2 \tag{87}$$

with g_1 = `rt_us_level`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.5 Credit

C.5.1 `cr_ig` — IG Credit

Hierarchy level 1. Method `single`.

> Rates-hedged by regression, not by analytic duration match.

Construction

$$f_t = x_t^{\text{LQD}} \tag{88}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. LQD is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{LQD}	`LQD`	instrument return: log total return of LQD
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\,\tilde{f}_t^{\text{rt_us_level}} - \hat{b}_{2,\tau}\,\tilde{f}_t^{\text{rt_us_slope}} \tag{89}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j\, \tilde{f}_s^{g_j} \right)^2 \tag{90}$$

with g_1, g_2 = `rt_us_level`, `rt_us_slope`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.5.2 `cr_hy` — High Yield

Hierarchy level 1. Method `single`.

> Also residualised against equity: HY beta to equity is large and leaving it in would double-count equity risk.

Construction

$$f_t = x_t^{\text{HYG}} \tag{91}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. HYG is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{HYG}	`HYG`	instrument return: log total return of HYG
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\,\tilde{f}_t^{\text{rt_us_level}} - \hat{b}_{2,\tau}\,\tilde{f}_t^{\text{rt_us_slope}} - \hat{b}_{3,\tau}\,\tilde{f}_t^{\text{eq_global}} \tag{92}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j\, \tilde{f}_s^{g_j} \right)^2 \tag{93}$$

with $g_1, \ldots, g_3$ = `rt_us_level`, `rt_us_slope`, `eq_global`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.5.3 `cr_loans` — Leveraged Loans

Hierarchy level 1. Method `single`.

> Floating rate, so little duration; the interesting part is what it adds over HY.

Construction

$$f_t = x_t^{\text{BKLN}} \tag{94}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. BKLN is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{BKLN}	`BKLN`	instrument return: log total return of BKLN
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\,\tilde{f}_t^{\text{rt_us_level}} - \hat{b}_{2,\tau}\,\tilde{f}_t^{\text{cr_hy}} \tag{95}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j\, \tilde{f}_s^{g_j} \right)^2 \tag{96}$$

with g_1, g_2 = `rt_us_level`, `cr_hy`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.5.4 `cr_em_hard` — EM Hard Currency

Hierarchy level 1. Method `single`.

> Residualised against equity as well: EM sovereign spreads carry real equity beta (0.68 before this was removed), which would otherwise be double-counted against the equity block.

Construction

$$f_t = x_t^{\text{EMB}} \tag{97}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. EMB is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{EMB}	`EMB`	instrument return: log total return of EMB
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\,\tilde{f}_t^{\text{rt_us_level}} - \hat{b}_{2,\tau}\,\tilde{f}_t^{\text{rt_us_slope}} - \hat{b}_{3,\tau}\,\tilde{f}_t^{\text{cr_hy}} - \hat{b}_{4,\tau}\,\tilde{f}_t^{\text{eq_global}} \tag{98}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j\,\tilde{f}_s^{g_j} \right)^2 \tag{99}$$

with $g_1, \ldots, g_4$ = `rt_us_level`, `rt_us_slope`, `cr_hy`, `eq_global`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.5.5 `cr_em_local` — EM Local Currency

Hierarchy level 2. Method `single`.

Residualised against USD as well: the local-currency leg is mostly FX.

Construction

$$f_t = x_t^{\text{EMLC}} \tag{100}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. EMLC is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{EMLC}	`EMLC`	instrument return: log total return of EMLC
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\,\tilde{f}_t^{\text{rt_us_level}} - \hat{b}_{2,\tau}\,\tilde{f}_t^{\text{cr_em_hard}} - \hat{b}_{3,\tau}\,\tilde{f}_t^{\text{fx_usd}} \tag{101}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j\,\tilde{f}_s^{g_j} \right)^2 \tag{102}$$

with $g_1, \ldots, g_3$ = `rt_us_level`, `cr_em_hard`, `fx_usd`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.6 FX

C.6.1 `fx_usd` — USD Broad

Hierarchy level 0. Method `single`.

Dollar index return. Not a funded position, so no cash leg.

Construction

$$f_t = r_t^{\text{DX-Y.NYB}} \tag{103}$$

1. No cash leg: this is not a funded position, so there is nothing to fund.

Symbol	Source	Meaning
$r_t^{\text{DX-Y.NYB}}$	`DX-Y.NYB`	instrument return: log total return of DX-Y.NYB

Orthogonalisation

$$\tilde{f}_t = f_t \tag{104}$$

Nothing sits above this factor in the hierarchy, so the raw and orthogonalised panels hold the same series.

C.6.2 `fx_jpy` — JPY (ex-USD)

Hierarchy level 1. Method `single`.

Sign flipped so a positive value means a stronger yen, the usual risk-off direction.

Construction

$$f_t = -1\, r_t^{\text{USDJPY=X}} \tag{105}$$

1. No cash leg: this is not a funded position, so there is nothing to fund.
2. Multiply by -1 so the factor points in the stated direction.

Symbol	Source	Meaning
$r_t^{\text{USDJPY=X}}$	`USDJPY=X`	instrument return: log total return of USDJPY=X

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\, \tilde{f}_t^{\text{fx_usd}} \tag{106}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j \tilde{f}_s^{g_j} \right)^2 \tag{107}$$

with g_1 = `fx_usd`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.6.3 `fx_em` — EM FX

Hierarchy level 1. Method `single`.

Construction

$$f_t = x_t^{\text{CEW}} \tag{108}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. CEW is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{CEW}	`CEW`	instrument return: log total return of CEW
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau} \tilde{f}_t^{\text{fx_usd}} \tag{109}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j \tilde{f}_s^{g_j} \right)^2 \tag{110}$$

with g_1 = `fx_usd`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.6.4 `fx_carry` — FX Carry (G4 approximation)

Hierarchy level 1. Method `fx_carry`.

> APPROXIMATION. Built from policy-rate differentials because the warehouse has no forward points, and limited to the three non-USD policy rates available. Narrower and noisier than a real G10 carry basket; treat its loading with corresponding scepticism. Also residualised against JPY: with only three legs and the yen as the perennial funding currency, the raw basket correlated -0.87 with fx_jpy, i.e. it was the yen short wearing a different name.

Construction

$$\kappa_{j,t} = \frac{i_{j,t-1} - i_{\mathrm{USD},t-1}}{100 \cdot 252}, \qquad p_{j,t} = \operatorname{sign}(\kappa_{j,t}) \tag{111}$$

$$f_t = \frac{1}{3} \sum_j p_{j,t} \left(s_j \, r_t^{X_j} + \kappa_{j,t} \right) \tag{112}$$

1. Carry is approximated from policy-rate differentials (CHF, EUR, JPY against USD) because the warehouse holds no forward points. Covered interest parity says the forward discount equals the rate differential, so this stands in for it.
2. The differential is lagged one day: the carry earned over day t is fixed by the rates set at $t - 1$.
3. Each leg is held long when it yields more than the dollar and short when it yields less, and the basket is the equally weighted average of the legs.
4. APPROXIMATION. Three non-USD legs is narrower and noisier than a real G10 carry basket; treat its loading with corresponding scepticism.

Symbol	Source	Meaning
$r_t^{\mathrm{USDCHF=X}}$	`USDCHF=X`	CHF spot leg, sign flipped so the quoted pair reads as long the foreign currency against the dollar: log total return of USDCHF=X
SNB:POLICY_RATE	`SNB:POLICY_RATE`	CHF policy rate i_{CHF}, annualised percent
$r_t^{\mathrm{EURUSD=X}}$	`EURUSD=X`	EUR spot leg: log total return of EURUSD=X
ECB:DFR	`ECB:DFR`	EUR policy rate i_{EUR}, annualised percent
$r_t^{\mathrm{USDJPY=X}}$	`USDJPY=X`	JPY spot leg, sign flipped so the quoted pair reads as long the foreign currency against the dollar: log total return of USDJPY=X
BOJ:IR01_OCRT	`BOJ:IR01_OCRT`	JPY policy rate i_{JPY}, annualised percent
FRED:DFF	`FRED:DFF`	USD policy rate, the funding leg

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau} \, \tilde{f}_t^{\mathrm{fx_usd}} - \hat{b}_{2,\tau} \, \tilde{f}_t^{\mathrm{fx_jpy}} \tag{113}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j \, \tilde{f}_s^{g_j} \right)^2 \tag{114}$$

with g_1, g_2 = `fx_usd`, `fx_jpy`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.7 Commodities

C.7.1 `cm_broad` — Broad Commodity

Hierarchy level 0. Method `single`.

DBC is a total-return futures fund: roll and collateral included.

Construction

$$f_t = x_t^{\text{DBC}} \tag{115}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. DBC is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{DBC}	`DBC`	instrument return: log total return of DBC
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t \tag{116}$$

Nothing sits above this factor in the hierarchy, so the raw and orthogonalised panels hold the same series.

C.7.2 `cm_energy` — Energy (ex-broad)

Hierarchy level 1. Method `single`.

Construction

$$f_t = x_t^{\text{USO}} \tag{117}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. USO is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{USO}	`USO`	instrument return: log total return of USO
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\, \tilde{f}_t^{\text{cm_broad}} \tag{118}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j \tilde{f}_s^{g_j} \right)^2 \tag{119}$$

with g_1 = `cm_broad`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.7.3 `cm_gold` — Gold (ex-broad)

Hierarchy level 1. Method `single`.

Construction

$$f_t = x_t^{\text{GLD}} \tag{120}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. GLD is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{GLD}	`GLD`	instrument return: log total return of GLD
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau} \tilde{f}_t^{\text{cm_broad}} \tag{121}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j \tilde{f}_s^{g_j} \right)^2 \tag{122}$$

with g_1 = `cm_broad`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.7.4 `cm_industrial` — Industrial Metals (ex-broad)

Hierarchy level 1. Method `single`.

Construction

$$f_t = x_t^{\text{CPER}} \tag{123}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. CPER is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{CPER}	`CPER`	instrument return: log total return of CPER
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\,\tilde{f}_t^{\text{cm_broad}} \tag{124}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j\, \tilde{f}_s^{g_j} \right)^2 \tag{125}$$

with g_1 = `cm_broad`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.7.5 `cm_agri` — Agriculture (ex-broad)

Hierarchy level 1. Method `single`.

Construction

$$f_t = x_t^{\text{DBA}} \tag{126}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. DBA is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{DBA}	`DBA`	instrument return: log total return of DBA
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\,\tilde{f}_t^{\text{cm_broad}} \tag{127}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j\, \tilde{f}_s^{g_j} \right)^2 \tag{128}$$

with g_1 = `cm_broad`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.8 Volatility

C.8.1 `vol_equity` — Equity Volatility (VIX futures)

Hierarchy level 1. Method `single`.

Long VIX-futures strategy return, residualised against equity.

Construction

$$f_t = x_t^{\text{VIXY}} \tag{129}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. VIXY is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{VIXY}	`VIXY`	instrument return: log total return of VIXY
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\, \tilde{f}_t^{\text{eq_global}} \tag{130}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j \tilde{f}_s^{g_j} \right)^2 \tag{131}$$

with g_1 = `eq_global`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.8.2 `vol_variance_premium` — Variance Risk Premium

Hierarchy level 1. Method `variance_premium`.

Daily payoff of a short variance position: yesterday's implied variance minus today's realised squared return.

Construction

$$f_t = \frac{1}{252} \left(\frac{\text{VIX}_{t-1}}{100} \right)^2 - \left(r_t^{\text{SPY}} \right)^2 \tag{132}$$

$$f_t \leftarrow k\, f_t, \qquad k = \frac{0.1/\sqrt{252}}{\text{sd}(f)} \tag{133}$$

1. Yesterday's implied variance is what a variance swap struck at $t-1$ pays against, so the lag is the contract and not a modelling choice.
2. Today's realised variance is the squared SPY return, which is what the daily leg of such a swap settles on.
3. The difference is the daily payoff of a short variance position, positive on average because implied sits above realised.
4. The payoff is convex in the underlying, so a linear beta on it is a first approximation and nothing more.
5. Rescale to 10% annualised volatility. The transform above yields a z-score, and every factor in this model is a return.

Symbol	Source	Meaning
FRED:VIXCLS	`FRED:VIXCLS`	VIX close in index points; divided by 100 it is an annualised volatility in decimal
r_t^{SPY}	`SPY`	underlying whose realised variance is delivered: log total return of SPY

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\, \tilde{f}_t^{\mathrm{eq_global}} \tag{134}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \Big(f_s - a - \sum_j b_j \tilde{f}_s^{g_j} \Big)^2 \tag{135}$$

with g_1 = `eq_global`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.8.3 `vol_rates` — Rates Volatility (surprise)

Hierarchy level 1. Method `abs_return_surprise`.

> Substitute for ^TYVIX, which CBOE discontinued in 2020: TLT's absolute return over yesterday's EWMA volatility forecast. Replaced the change in a 21-day rolling volatility, half of whose variance was the squared return leaving the window three weeks earlier.

Construction

$$f_t = \frac{|r_t^{\mathrm{TLT}}|}{s_{t-1}} - \sqrt{2/\pi} \tag{136}$$

$$s_t^2 = \lambda\, s_{t-1}^2 + (1-\lambda)\big(r_t^{\mathrm{TLT}}\big)^2, \qquad \lambda = 0.5^{1/21} \tag{137}$$

$$f_t \leftarrow k\, f_t, \qquad k = \frac{0.1/\sqrt{252}}{\mathrm{sd}(f)} \tag{138}$$

1. Divide today's absolute return by yesterday's forecast. Nothing about day t enters the forecast it is measured against.
2. Subtract $\sqrt{2/\pi}$, the mean absolute value of a standard normal, so the factor is zero on average when the forecast is right and positive on a day more volatile than expected.
3. A day's return enters once, on its own day, and fades with the average thereafter. The change in a 21-day rolling volatility it replaced moved by the squared return entering the window minus the one leaving it, so half its variance was an echo of a day three weeks earlier.
4. Dividing by the forecast keeps the scale from following the level of volatility, so one crisis day does not dominate a covariance estimated on it.
5. Undefined until TLT has 21 returns behind it.
6. This is a volatility shock, not a tradable return. It stands in for a rates-volatility series because CBOE discontinued ^TYVIX in 2020.
7. Rescale to 10% annualised volatility. The transform above yields a z-score, and every factor in this model is a return.

Symbol	Source	Meaning
r_t^{TLT}	`TLT`	instrument whose volatility is tracked: log total return of TLT

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\, \tilde{f}_t^{\mathrm{rt_us_level}} \tag{139}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \Big(f_s - a - \sum_j b_j\, \tilde{f}_s^{g_j} \Big)^2 \tag{140}$$

with g_1 = `rt_us_level`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.9 Liquidity & Stress

C.9.1 `liq_funding` — Funding Spread (SOFR-EFFR)

Hierarchy level 0. Method `spread_level`.

> TED and LIBOR-OIS are discontinued; SOFR over the effective funds rate is the live equivalent. Standardised because basis points are not comparable to the return-scaled factors.

Construction

$$s_t = \text{FRED:SOFR}_t - \text{FRED:EFFR}_t \tag{141}$$

$$f_t = \frac{\Delta s_t}{\mathrm{sd}\left(\Delta s_{t-251}, \ldots, \Delta s_t\right)} \tag{142}$$

$$f_t \leftarrow k\, f_t, \qquad k = \frac{0.1/\sqrt{252}}{\mathrm{sd}(f)} \tag{143}$$

1. Form the spread between the two quoted levels.
2. Divide the daily change by its own trailing volatility. Basis points are not comparable with the return-scaled blocks, and the scaling window is strictly trailing: a full-sample standard deviation would leak future volatility into every historical observation.
3. Rescale to 10% annualised volatility. The transform above yields a z-score, and every factor in this model is a return.

Symbol	Source	Meaning
FRED:SOFR	`FRED:SOFR`	first leg of the spread, in percent
FRED:EFFR	`FRED:EFFR`	second leg of the spread, in percent

Orthogonalisation

$$\tilde{f}_t = f_t \tag{144}$$

Nothing sits above this factor in the hierarchy, so the raw and orthogonalised panels hold the same series.

C.9.2 `liq_conditions` — Financial Conditions (NFCI)

Hierarchy level 0. Method `level_transform`.

> Weekly series; the daily factor is zero between releases rather than forward-filled, since a forward fill would manufacture information. Each reading is the first release, on the Wednesday it was published - not the Friday of the week it describes, which FRED dates it by and which is five days before anyone knew it.

Construction

$$f_t = \begin{cases} \dfrac{\Delta s_k}{\mathrm{sd}(\Delta s_{k-51}, \ldots, \Delta s_k)} & t = \tau_k, \text{ a release day} \\ 0 & \text{otherwise} \end{cases} \tag{145}$$

$$f_t \leftarrow k\, f_t, \qquad k = \frac{0.1/\sqrt{252}}{\mathrm{sd}(f)} \tag{146}$$

1. The series prints weekly, not daily.
2. Each reading is the first release, placed on the day it was published (rolled to the next trading day over a holiday), not on the reference date the source files it under. A change is the difference between consecutive readings as published, so it includes the revision to the previous week: that, too, was news on the day.
3. It is NOT forward filled. A filled series carries no new information between releases, which manufactures autocorrelation, understates standard errors and can smuggle in lookahead.
4. Instead the standardised change lands on the publication day and the factor is exactly zero in between: a release-event factor, sparse daily.
5. The scaling window counts releases rather than days, so 52 of them is about a year of the series' own history.
6. Rescale to 10% annualised volatility. The transform above yields a z-score, and every factor in this model is a return.

Symbol	Source	Meaning
FRED:NFCI@release	`FRED:NFCI@release`	the published level series

Orthogonalisation

$$\tilde{f}_t = f_t \tag{147}$$

Nothing sits above this factor in the hierarchy, so the raw and orthogonalised panels hold the same series.

C.9.3 `liq_risk_off` — Cross-Asset Risk-Off

Hierarchy level 2. Method `basket`.

> The residual co-movement of classic haven and risk assets after each leg's own block exposure is removed.

Construction

$$f_t = \frac{r_t^{\mathrm{TLT}} + r_t^{\mathrm{UUP}} + r_t^{\mathrm{GLD}}}{3} - \frac{r_t^{\mathrm{HYG}} + r_t^{\mathrm{EEM}}}{2} \tag{148}$$

1. Equally weighted long basket of 3: each leg carries $1/3$ of the notional.
2. Minus an equally weighted short basket of 2. Both sides carry the market direction, so it cancels and what survives is the spread between them.

3. Self-financing, so no cash leg.

Symbol	Source	Meaning
r_t^{TLT}	`TLT`	long leg: log total return of TLT
r_t^{UUP}	`UUP`	long leg: log total return of UUP
r_t^{GLD}	`GLD`	long leg: log total return of GLD
r_t^{HYG}	`HYG`	short leg: log total return of HYG
r_t^{EEM}	`EEM`	short leg: log total return of EEM

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\,\tilde{f}_t^{\mathrm{eq_global}} - \hat{b}_{2,\tau}\,\tilde{f}_t^{\mathrm{rt_us_level}} - \hat{b}_{3,\tau}\,\tilde{f}_t^{\mathrm{cr_hy}} \tag{149}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \Big(f_s - a - \sum_j b_j\, \tilde{f}_s^{g_j} \Big)^2 \tag{150}$$

with $g_1, \ldots, g_3$ = `eq_global`, `rt_us_level`, `cr_hy`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.10 Alternative Risk Premia

C.10.1 `arp_trend` — Time-Series Momentum

Hierarchy level 3. Method `tsmom`.

> 12-month-minus-1 signal, inverse-volatility sized, equally weighted across assets. Validated against DBMF and KMLM over their short lives.

Construction

$$S_{i,t} = \operatorname{sign}\Big(\sum_{u=t-252}^{t-22} r_{i,u} \Big), \qquad v_{i,t} = \operatorname{sd}\big(r_{i,t-63}, \ldots, r_{i,t-1} \big) \tag{151}$$

$$w_{i,t} = \frac{S_{i,t}}{v_{i,t}}, \qquad f_t = \frac{\sum_i w_{i,t}\, r_{i,t}}{\sum_i |w_{i,t}|} \tag{152}$$

1. The signal at t reads returns through t-22 only, and the position it implies is applied to the return at t. Nothing about day t enters its own weight.
2. Skipping the most recent 21 days is the standard 12-minus-1 construction, which keeps short-term reversal out of the trend signal.
3. Inverse-volatility sizing puts each of the 13 instruments on a comparable risk footing; without it the equity legs would drown out the bond legs.

4. Dividing by gross exposure makes the factor a return on one unit of notional rather than a sum that grows with the size of the universe.

Symbol	Source	Meaning
r_t^{SPY}	`SPY`	instrument return: log total return of SPY
r_t^{EFA}	`EFA`	instrument return: log total return of EFA
r_t^{EEM}	`EEM`	instrument return: log total return of EEM
r_t^{TLT}	`TLT`	instrument return: log total return of TLT
r_t^{IEF}	`IEF`	instrument return: log total return of IEF
r_t^{DBC}	`DBC`	instrument return: log total return of DBC
r_t^{GLD}	`GLD`	instrument return: log total return of GLD
r_t^{USO}	`USO`	instrument return: log total return of USO
r_t^{HYG}	`HYG`	instrument return: log total return of HYG
r_t^{LQD}	`LQD`	instrument return: log total return of LQD
r_t^{UUP}	`UUP`	instrument return: log total return of UUP
r_t^{FXE}	`FXE`	instrument return: log total return of FXE
r_t^{FXY}	`FXY`	instrument return: log total return of FXY

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\,\tilde{f}_t^{\text{eq_global}} - \hat{b}_{2,\tau}\,\tilde{f}_t^{\text{rt_us_level}} - \hat{b}_{3,\tau}\,\tilde{f}_t^{\text{cm_broad}} - \hat{b}_{4,\tau}\,\tilde{f}_t^{\text{fx_usd}} \tag{153}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \Big(f_s - a - \sum_j b_j\, \tilde{f}_s^{g_j} \Big)^2 \tag{154}$$

with $g_1, \ldots, g_4$ = `eq_global`, `rt_us_level`, `cm_broad`, `fx_usd`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.10.2 `arp_xs_momentum` — Cross-Sectional Momentum

Hierarchy level 3. Method `xs_momentum`.

> Long the top third, short the bottom third by trailing return; dollar-neutral by construction.

Construction

$$M_{i,t} = \sum_{u=t-252}^{t-22} r_{i,u}, \qquad q_{i,t} = \mathrm{rank}_{\mathrm{pct}}\big(M_{\cdot,t}\big)_i \tag{155}$$

$$w_{i,t} = \frac{\mathbf{1}\{q_{i,t} > 2/3\}}{n_t^{\text{long}}} - \frac{\mathbf{1}\{q_{i,t} < 1/3\}}{n_t^{\text{short}}}, \qquad f_t = \sum_i w_{i,t}\, r_{i,t} \tag{156}$$

1. Rank the 13 instruments by trailing return, again skipping the most recent 21 days.
2. Long the top third, short the bottom third, equally weighted within each leg.
3. Dollar-neutral by construction: the long and the short weights each sum to one, so the common market direction cancels and what is left is the spread between winners and losers.

Symbol	Source	Meaning
r_t^{SPY}	`SPY`	instrument return: log total return of SPY
r_t^{EFA}	`EFA`	instrument return: log total return of EFA
r_t^{EEM}	`EEM`	instrument return: log total return of EEM
r_t^{TLT}	`TLT`	instrument return: log total return of TLT
r_t^{IEF}	`IEF`	instrument return: log total return of IEF
r_t^{DBC}	`DBC`	instrument return: log total return of DBC
r_t^{GLD}	`GLD`	instrument return: log total return of GLD
r_t^{USO}	`USO`	instrument return: log total return of USO
r_t^{HYG}	`HYG`	instrument return: log total return of HYG
r_t^{LQD}	`LQD`	instrument return: log total return of LQD
r_t^{UUP}	`UUP`	instrument return: log total return of UUP
r_t^{FXE}	`FXE`	instrument return: log total return of FXE
r_t^{FXY}	`FXY`	instrument return: log total return of FXY

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\, \tilde{f}_t^{\mathrm{eq_global}} - \hat{b}_{2,\tau}\, \tilde{f}_t^{\mathrm{arp_trend}} \tag{157}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \Big(f_s - a - \sum_j b_j\, \tilde{f}_s^{g_j} \Big)^2 \tag{158}$$

with g_1, g_2 = `eq_global`, `arp_trend`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.11 Equity Sector (role 'sector': a single stock adds only its own)

C.11.1 `sec_tech` — Sector: Information Technology

Hierarchy level 2. Method `single`.

> XLK total return less cash, residualised against the market and the cyclical tilt. Enters a regression only for a Information Technology stock, and only when the model agent finds it earns its place.

Construction

$$f_t = x_t^{\mathrm{XLK}} \tag{159}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. XLK is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{XLK}	`XLK`	instrument return: log total return of XLK
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\,\tilde{f}_t^{\text{eq_global}} - \hat{b}_{2,\tau}\,\tilde{f}_t^{\text{eq_us}} - \hat{b}_{3,\tau}\,\tilde{f}_t^{\text{eq_cyclical}} \tag{160}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j\, \tilde{f}_s^{g_j} \right)^2 \tag{161}$$

with $g_1, \ldots, g_3$ = `eq_global`, `eq_us`, `eq_cyclical`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.11.2 `sec_financials` — Sector: Financials

Hierarchy level 2. Method `single`.

> XLF total return less cash, residualised against the market and the cyclical tilt. Enters a regression only for a Financials stock, and only when the model agent finds it earns its place.

Construction

$$f_t = x_t^{\text{XLF}} \tag{162}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. XLF is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{XLF}	`XLF`	instrument return: log total return of XLF
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\,\tilde{f}_t^{\text{eq_global}} - \hat{b}_{2,\tau}\,\tilde{f}_t^{\text{eq_us}} - \hat{b}_{3,\tau}\,\tilde{f}_t^{\text{eq_cyclical}} \tag{163}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j \tilde{f}_s^{g_j} \right)^2 \tag{164}$$

with $g_1, \ldots, g_3$ = `eq_global`, `eq_us`, `eq_cyclical`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.11.3 `sec_health` — Sector: Health Care

Hierarchy level 2. Method `single`.

> XLV total return less cash, residualised against the market and the cyclical tilt. Enters a regression only for a Health Care stock, and only when the model agent finds it earns its place.

Construction

$$f_t = x_t^{\text{XLV}} \tag{165}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. XLV is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{XLV}	`XLV`	instrument return: log total return of XLV
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\, \tilde{f}_t^{\text{eq_global}} - \hat{b}_{2,\tau}\, \tilde{f}_t^{\text{eq_us}} - \hat{b}_{3,\tau}\, \tilde{f}_t^{\text{eq_cyclical}} \tag{166}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j \tilde{f}_s^{g_j} \right)^2 \tag{167}$$

with $g_1, \ldots, g_3$ = `eq_global`, `eq_us`, `eq_cyclical`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.11.4 `sec_industrials` — Sector: Industrials

Hierarchy level 2. Method `single`.

> XLI total return less cash, residualised against the market and the cyclical tilt. Enters a regression only for a Industrials stock, and only when the model agent finds it earns its place.

Construction

$$f_t = x_t^{\text{XLI}} \tag{168}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. XLI is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{XLI}	`XLI`	instrument return: log total return of XLI
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\,\tilde{f}_t^{\text{eq_global}} - \hat{b}_{2,\tau}\,\tilde{f}_t^{\text{eq_us}} - \hat{b}_{3,\tau}\,\tilde{f}_t^{\text{eq_cyclical}} \tag{169}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j\, \tilde{f}_s^{g_j} \right)^2 \tag{170}$$

with $g_1, \ldots, g_3$ = `eq_global`, `eq_us`, `eq_cyclical`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.11.5 `sec_discretionary` — Sector: Consumer Discretionary

Hierarchy level 2. Method `single`.

> XLY total return less cash, residualised against the market and the cyclical tilt. Enters a regression only for a Consumer Discretionary stock, and only when the model agent finds it earns its place.

Construction

$$f_t = x_t^{\text{XLY}} \tag{171}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. XLY is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{XLY}	`XLY`	instrument return: log total return of XLY
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\,\tilde{f}_t^{\text{eq_global}} - \hat{b}_{2,\tau}\,\tilde{f}_t^{\text{eq_us}} - \hat{b}_{3,\tau}\,\tilde{f}_t^{\text{eq_cyclical}} \tag{172}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j\,\tilde{f}_s^{g_j} \right)^2 \tag{173}$$

with $g_1, \ldots, g_3$ = `eq_global`, `eq_us`, `eq_cyclical`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.11.6 `sec_staples` — Sector: Consumer Staples

Hierarchy level 2. Method `single`.

> XLP total return less cash, residualised against the market and the cyclical tilt. Enters a regression only for a Consumer Staples stock, and only when the model agent finds it earns its place.

Construction

$$f_t = x_t^{\text{XLP}} \tag{174}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. XLP is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{XLP}	`XLP`	instrument return: log total return of XLP
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\,\tilde{f}_t^{\text{eq_global}} - \hat{b}_{2,\tau}\,\tilde{f}_t^{\text{eq_us}} - \hat{b}_{3,\tau}\,\tilde{f}_t^{\text{eq_cyclical}} \tag{175}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j\,\tilde{f}_s^{g_j} \right)^2 \tag{176}$$

with $g_1, \ldots, g_3$ = `eq_global`, `eq_us`, `eq_cyclical`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.11.7 `sec_energy` — Sector: Energy

Hierarchy level 2. Method `single`.

> XLE total return less cash, residualised against the market and the cyclical tilt. Enters a regression only for a Energy stock, and only when the model agent finds it earns its place.

Construction

$$f_t = x_t^{\text{XLE}} \tag{177}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. XLE is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{XLE}	`XLE`	instrument return: log total return of XLE
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\,\tilde{f}_t^{\text{eq_global}} - \hat{b}_{2,\tau}\,\tilde{f}_t^{\text{eq_us}} - \hat{b}_{3,\tau}\,\tilde{f}_t^{\text{eq_cyclical}} \tag{178}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \Big(f_s - a - \sum_j b_j\, \tilde{f}_s^{g_j} \Big)^2 \tag{179}$$

with $g_1, \ldots, g_3$ = `eq_global`, `eq_us`, `eq_cyclical`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.11.8 `sec_materials` — Sector: Materials

Hierarchy level 2. Method `single`.

> XLB total return less cash, residualised against the market and the cyclical tilt. Enters a regression only for a Materials stock, and only when the model agent finds it earns its place.

Construction

$$f_t = x_t^{\text{XLB}} \tag{180}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. XLB is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{XLB}	`XLB`	instrument return: log total return of XLB
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\,\tilde{f}_t^{\text{eq_global}} - \hat{b}_{2,\tau}\,\tilde{f}_t^{\text{eq_us}} - \hat{b}_{3,\tau}\,\tilde{f}_t^{\text{eq_cyclical}} \tag{181}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j\, \tilde{f}_s^{g_j} \right)^2 \tag{182}$$

with $g_1, \ldots, g_3$ = `eq_global`, `eq_us`, `eq_cyclical`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.11.9 `sec_utilities` — Sector: Utilities

Hierarchy level 2. Method `single`.

> XLU total return less cash, residualised against the market and the cyclical tilt. Enters a regression only for a Utilities stock, and only when the model agent finds it earns its place.

Construction

$$f_t = x_t^{\text{XLU}} \tag{183}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. XLU is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{XLU}	`XLU`	instrument return: log total return of XLU
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\,\tilde{f}_t^{\text{eq_global}} - \hat{b}_{2,\tau}\,\tilde{f}_t^{\text{eq_us}} - \hat{b}_{3,\tau}\,\tilde{f}_t^{\text{eq_cyclical}} \tag{184}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j\, \tilde{f}_s^{g_j} \right)^2 \tag{185}$$

with $g_1, \ldots, g_3$ = `eq_global`, `eq_us`, `eq_cyclical`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.11.10 `sec_real_estate` — Sector: Real Estate

Hierarchy level 2. Method `single`.

> XLRE total return less cash, residualised against the market and the cyclical tilt. Enters a regression only for a Real Estate stock, and only when the model agent finds it earns its place. XLRE starts in October 2015.

Construction

$$f_t = x_t^{\text{XLRE}} \tag{186}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. XLRE is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{XLRE}	`XLRE`	instrument return: log total return of XLRE
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\,\tilde{f}_t^{\text{eq_global}} - \hat{b}_{2,\tau}\,\tilde{f}_t^{\text{eq_us}} - \hat{b}_{3,\tau}\,\tilde{f}_t^{\text{eq_cyclical}} \tag{187}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j \tilde{f}_s^{g_j} \right)^2 \tag{188}$$

with $g_1, \ldots, g_3$ = `eq_global`, `eq_us`, `eq_cyclical`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.11.11 `sec_communication` — Sector: Communication Services

Hierarchy level 2. Method `single`.

> XLC total return less cash, residualised against the market and the cyclical tilt. Enters a regression only for a Communication Services stock, and only when the model agent finds it earns its place. XLC starts in June 2018.

Construction

$$f_t = x_t^{\text{XLC}} \tag{189}$$

1. Subtract the overnight rate: $x_t = r_t - c_{t-1}$. XLC is a funded long position, so its return has to be measured over cash.

Symbol	Source	Meaning
r_t^{XLC}	`XLC`	instrument return: log total return of XLC
c_t	`FRED:DFF`	daily cash rate: FRED:DFF$_t$/100/252, lagged one day because the overnight rate earned on day t is set at $t-1$

Orthogonalisation

$$\tilde{f}_t = f_t - \hat{a}_\tau - \hat{b}_{1,\tau}\,\tilde{f}_t^{\text{eq_global}} - \hat{b}_{2,\tau}\,\tilde{f}_t^{\text{eq_us}} - \hat{b}_{3,\tau}\,\tilde{f}_t^{\text{eq_cyclical}} \tag{190}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j\,\tilde{f}_s^{g_j} \right)^2 \tag{191}$$

with $g_1, \ldots, g_3$ = `eq_global`, `eq_us`, `eq_cyclical`, each taken as its own *orthogonalised* series, and τ the most recent refit at or before t.

C.12 The orthogonalisation, once

The equation above is the same for every factor that has targets; only the regressor list changes. In general form, for k targets $g_1, \ldots, g_k$:

$$\tilde{f}_t = f_t - \hat{a}_\tau - \sum_{j=1}^{k} \hat{b}_{j,\tau}\,\tilde{f}_t^{g_j} \tag{192}$$

$$(\hat{a}_\tau, \hat{b}_\tau) = \arg\min_{a,b} \sum_{s=\tau-504}^{\tau-1} \left(f_s - a - \sum_j b_j\,\tilde{f}_s^{g_j} \right)^2 \tag{193}$$

1. Coefficients come from ordinary least squares on the trailing 504 observations (2 years), refitted every 21 observations and applied forward until the next refit.
2. The fitting window ends at $\tau - 1$, so day t is never part of the regression that residualises it. That is the whole reason for the rolling refit: a full-sample residual would put future information into historical factor values, and this project's headline output is a predicted-versus-realised risk comparison, which such a leak would flatter exactly where the model is being judged.
3. The regressors are the targets' own orthogonalised series, so the hierarchy compounds: by the time a level-3 factor is residualised, the level-0 factors it sees have already had everything above them removed.
4. Before the first refit, with fewer than 252 usable observations or a missing regressor on the day, the orthogonalised value is NULL rather than a silent fallback to the raw value. A factor that quietly stops being orthogonal on some dates is worse than one with a documented gap.

Two alternative modes exist and are not used. `expanding` fits on all history to date: causal, but it never forgets, so a loading that was high in one crisis stays high for a decade and the residual acquires a systematic negative loading on its own regressor. `full_sample` fits once on everything: exactly orthogonal, and what commercial risk models do, but every historical factor value then contains information from its own future. Measured residual correlation of `eq_em` against `eq_global` over 2009–2026: rolling -0.05, expanding -0.34, full sample 0.00.

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